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Modelling Wellbeing: Comparing Statistical, Machine Learning, Hybrid and Bayesian Network Models on Psychological Data

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Abstract

Psychological wellbeing is shaped by a complex interplay of personal, social, economic, and environmental factors operating at both individual and societal levels. This thesis presents a systematic comparative study of statistical, machine learning, hybrid, and Bayesian Network models for predicting and explaining wellbeing outcomes across four datasets spanning two levels of analysis: workplace mental health, student depression, Canadian life satisfaction, and the World Happiness Report Gallup Poll dataset covering 166 countries across 2005 to 2020.

Results show that predictive performance is driven primarily by data quality and signal concentration rather than algorithm choice. Ensemble methods achieved the strongest performance across datasets, with the macro-level Gallup dataset reaching a cross-validated R^2 of 0.850 compared to 0.422 for the best individual-level regression task. Hybrid models failed to outperform well-configured baselines across all datasets. Bayesian Network structure learning revealed between 11 and 79 percent causally implausible edges in unconstrained graphs, confirming that data-driven causal discovery requires domain constraints to be reliable. Expert-constrained Bayesian Networks recovered theoretically consistent causal pathways at negligible statistical cost, including academic pressure and financial stress as direct causes of student depression and GDP per capita as the sole direct parent of national happiness. SHAP analysis confirmed directional consistency across model families and identified proxy predictors whose removal improved causal interpretability while preserving predictive performance.

These findings demonstrate that machine learning and Bayesian Network models serve complementary rather than competing purposes: machine learning optimises predictive accuracy while Bayesian Networks recover interpretable causal structure, contributing empirical evidence to the prediction-explanation trade-off debate in psychological science.

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Chapter 1 - Introduction

1.1. Motivation

Mental health and psychological wellbeing are among the most complex and consequential outcomes in modern society. Conditions such as depression, chronic stress, and low life satisfaction affect hundreds of millions of people worldwide, with significant consequences for individuals, organisations, and public health systems [Helliwell2023]. Despite a broad scientific consensus on the factors that contribute to wellbeing - including social support, economic security, physical health, and psychological resilience [Diener1984; OECD2020] - the question of how accurately and interpretably these factors can be modelled using computational methods remains open [Breiman2001; Shmueli2010].

Two parallel challenges motivate this thesis. The first is the distinction between individual-level and societal-level data. Survey responses from employees or students capture rich personal experiences but are characterised by high noise, subjective bias, and complex interactions among predictors [Yarkoni2017; Shmueli2010]. Aggregate indicators such as GDP per capita, life expectancy, and institutional trust capture structural conditions at the country level but abstract away individual variation [Helliwell2023; OECD2020]. Whether predictive models perform differently across these two levels of analysis - and whether the same modeling frameworks can be applied consistently to both - is a question with both methodological and practical implications [Breiman2001].

The second challenge concerns the tension between prediction and explanation [Breiman2001; Shmueli2010]. Machine learning models have demonstrated strong predictive performance across a wide range of psychological and health outcomes [Hastie2009; Kuhn2013], but their internal logic is often opaque [Molnar2019]. Statistical models offer interpretability but may miss nonlinear relationships [Hastie2009]. Bayesian Networks occupy a distinctive position: they are generative probabilistic models that can encode causal structure explicitly, enabling reasoning under uncertainty in a way that neither regression nor ensemble methods can match [Pearl2009; Scutari2014]. Whether Bayesian Networks can recover plausible causal structures from observational wellbeing data - and whether those structures add value beyond what machine learning feature importance already reveals [Lundberg2017] - is the central methodological question of this thesis [Pearl2009, Scutari2014].

These challenges are not purely academic. Policymakers designing mental health interventions, employers evaluating workplace support programs, and researchers studying the determinants of happiness all need models that are not only accurate but interpretable and causally grounded. A model that predicts depression with high accuracy but cannot explain which factors drive the prediction, or in which direction, is of limited practical value for intervention design [Shmueli2010, Breiman2001].

1.2. Scope

This thesis presents a systematic comparative study of statistical, machine learning, hybrid, and Bayesian Network models for predicting and explaining psychological wellbeing under uncertainty, applied consistently across both individual-level and societal-level data.

The study is grounded in four datasets spanning two levels of analysis. At the individual level, three datasets are examined: a survey of mental health conditions in the technology industry, a survey of depression among students, and a Canadian population health survey measuring life satisfaction. At the societal level, the World Happiness Report Gallup Poll dataset - covering 166 countries across 2005–2020 - provides a macro-level comparison. An additional synthetic Kaggle dataset nominally derived from the World Happiness Report is included as a data quality case study, demonstrating the consequences of absent signal for all modelling approaches. All datasets were obtained from Kaggle, an open-access platform for public datasets [Kaggle].

The scope is deliberately comparative. The same modeling frameworks are applied consistently across all datasets, enabling direct cross-dataset and cross-model evaluation. The study does not aim to perform clinical diagnosis or psychological assessment. Its focus is on modelling self-reported mental health and wellbeing outcomes using different predictive approaches, with particular emphasis on the causal structure of the data and the interpretability of model outputs.

Wellbeing is operationalised across multiple constructs depending on the dataset: work interference and treatment-seeking behaviour in the workplace context [Diener1984], depression as a binary clinical indicator in the student context, life satisfaction as a continuous self-report measure in the Canadian survey [Pavot1993], and the Cantril Life Ladder score as a societal happiness measure in the Gallup dataset [Helliwell2023, OECD2020].

1.3. Research Questions

The study is organised around five research questions that together address the comparative performance, interpretability, causal validity, and robustness of the modelling frameworks examined:

RQ1. How do different modeling approaches perform when predicting wellbeing at individual versus societal levels? This question examines whether predictive performance - measured by classification metrics at the individual level and regression metrics at the societal level - differs systematically between data levels, and whether that difference varies across model families.

RQ2. How does interpretability differ between regression, machine learning, and Bayesian Networks? This question examines the qualitative and quantitative differences in how each model family explains its predictions, using coefficient analysis, SHAP values, feature importance rankings, and Bayesian Network conditional probability tables as complementary interpretability tools [Lundberg2017, Molnar2019, Goldstein2015].

RQ3. Are Bayesian Networks better at capturing plausible causal structures in wellbeing data? This question evaluates whether Bayesian Network structure learning, with and without expert constraints, recovers directed acyclic graphs whose edge directions are consistent with established domain knowledge about the causal determinants of mental health and wellbeing [Pearl2009, Morgan2015].

RQ4. Can Bayesian Networks reveal causal structures that ML models cannot? This question examines whether the explicit directional edges learned by Bayesian Networks provide causal information beyond what machine learning feature importance conveys, and what the conditions are under which data-driven structure learning succeeds or fails [Yarkoni2017, Scutari2014].

RQ5. How robust are these models to missing data, noise, and correlated predictors? This question evaluates model stability across cross-validation folds, sensitivity to the removal of dominant predictors, and performance under conditions of high missingness and correlated feature sets [Hastie2009, Kuhn2013].

1.4. Overview of Approach

The methodology follows a consistent pipeline applied to all datasets. Each dataset is preprocessed in two versions: a machine learning version with imputation, one-hot encoding, and missingness indicators, and a Bayesian Network version preserving missing values as discrete states for structure learning. Baseline machine learning models - logistic or linear regression, decision tree, random forest, and XGBoost - are trained and evaluated using stratified cross-validation with multiple metrics. Hybrid models combining nonlinear feature selection with linear estimators, and stacking ensembles, are evaluated against the baselines. Bayesian Networks are learned using Hill-Climb Search with BIC scoring, both unconstrained and with expert-defined forbidden edges encoding domain knowledge about causal ordering.

Interpretability is analysed at two levels. At the global level, SHAP TreeExplainer values provide model-agnostic feature importance and directional effect estimates for the best-performing model on each dataset. Sensitivity analyses remove dominant proxy predictors to assess robustness. At the structural level, Bayesian Network graphs are inspected for causally implausible edges, and BIC scores are compared between unconstrained and constrained versions to quantify the statistical cost of expert constraints.

Cross-dataset comparison is performed through unified performance tables, visualisations of hybrid model gain over baselines, and a Bayesian Network consolidation summarising structure quality, inference results, and causal pathway robustness across all datasets. The thesis concludes with a discussion connecting empirical findings to the five research questions and to the broader prediction-explanation debate in psychological science [Breiman2001, Shmueli2010].

Chapter 2 – Literature Review

2.1. Overview

This chapter situates the thesis within three intersecting bodies of literature: the debate between predictive and explanatory modelling in statistical and psychological science, the theory and practice of Bayesian Network causal discovery, and the measurement of subjective wellbeing at individual and societal levels. Together these three strands provide the conceptual foundation for the comparative modelling framework developed in this thesis and motivate the five research questions introduced in Chapter 1.

2.2. Prediction vs Explanation Debate

The relationship between statistical explanation and predictive accuracy has been a central tension in quantitative science for decades. The distinction was formalised most sharply by Breiman [Breiman2001], who identified two cultures in statistical modelling: the data modelling culture, which assumes a stochastic data-generating process and prioritises interpretable parametric models, and the algorithmic modelling culture, which treats the process as unknown and prioritises predictive accuracy. Breiman argued that psychology and the social sciences had become overly committed to the data modelling culture at the expense of predictive validity, and that algorithmic methods such as ensemble trees and neural networks offered substantial gains in out-of-sample performance.

Shmueli [Shmueli2010] developed this distinction more formally by introducing a framework for distinguishing explanatory modelling - aimed at testing causal hypotheses about the data-generating process - from predictive modelling - aimed at producing accurate forecasts of new observations. She argued that the two goals require different model selection criteria, different evaluation metrics, and different interpretive standards, and that conflating them leads to both poor predictions and poor causal inference. A model that fits the training data well and has theoretically grounded coefficients may nevertheless perform poorly on held-out data, while a model with strong out-of-sample performance may be causally uninterpretable. This framework directly motivates the comparative design of this thesis, which evaluates both predictive performance and causal interpretability across model families.

Yarkoni and Westfall [Yarkoni2017] extended this argument to psychological science, demonstrating that psychology's focus on causal explanation had produced a literature of intricate theories with limited predictive validity, and proposing that machine learning principles - regularisation, cross-validation, and out-of-sample evaluation - would produce more generalisable psychological models.

The growing adoption of complex machine learning models in applied settings has generated a parallel literature on post-hoc interpretability. Hastie, Tibshirani, and Friedman [Hastie2009] provide the statistical foundation for ensemble methods and the conditions under which feature importance measures are reliable. Lundberg and Lee [Lundberg2017] unified feature importance methods under the SHAP framework, providing a theoretically grounded decomposition of predictions into additive feature contributions based on Shapley values, satisfying axiomatic properties that earlier importance measures do not. Molnar [Molnar2019] situates SHAP alongside other explanation methods within a comprehensive taxonomy of interpretable machine learning.

In this thesis, SHAP is applied as the primary interpretability tool for the best-performing ML model on each dataset, complementing the structural interpretability provided by Bayesian Network graphs and coefficient tables from linear models.

2.3. Bayesian Networks Causal Discovery

Bayesian Networks are directed acyclic graphical models that represent a joint probability distribution over a set of variables through a product of conditional distributions [Scutari2014]. Each node represents a variable and each directed edge represents a conditional dependency, encoding the assertion that a child variable is directly influenced by its parent variables given the remaining network structure. The absence of an edge between two nodes encodes conditional independence, making Bayesian Networks a natural language for expressing causal hypotheses about data-generating processes.

Pearl [Pearl2009] established the theoretical foundations for causal inference in graphical models, introducing the do-calculus as a formal framework for reasoning about interventions - the effect of setting a variable to a fixed value - as distinct from observations. His framework distinguishes between associational quantities, which can be estimated from observational data, and causal quantities, which require either experimental manipulation or structural assumptions encoded in a directed acyclic graph. The d-separation criterion provides a graphical test for conditional independence, enabling the derivation of testable implications from a causal model without intervention data.

A fundamental limitation of data-driven structure learning is that it cannot distinguish between Markov equivalent graphs - directed acyclic graphs that encode the same conditional independence relationships but differ in edge direction. Without additional constraints derived from domain knowledge - such as the temporal ordering of variables or theoretical assumptions about causal direction - structure learning algorithms may recover statistically equivalent but causally implausible graphs. This limitation is directly relevant to the wellbeing context, where survey variables often measure co-occurring constructs whose causal ordering is theoretically motivated but not identifiable from observational data alone.

In this thesis, expert knowledge is incorporated into the structure learning process by specifying forbidden edges encoding known causal ordering constraints - for example, that demographic variables such as age and gender cannot be caused by psychological outcome variables, and that treatment-seeking behaviour cannot cause a family history of mental illness. The impact of these constraints on both statistical fit and causal plausibility is evaluated systematically across all datasets.

2.4. Wellbeing Measurement

Subjective wellbeing is a multidimensional construct encompassing cognitive evaluations of life quality, the presence of positive affect, and the absence of negative affect [Diener1984]. Diener's foundational review established subjective wellbeing as a legitimate scientific construct and identified its three primary components: life satisfaction as a cognitive judgement, positive affect as the frequency of pleasant emotional states, and negative affect as the frequency of unpleasant emotional states. This tripartite model has been the dominant framework in wellbeing research for four decades and provides the theoretical grounding for the outcome variables used in this thesis.

At the individual level, life satisfaction is most commonly measured using the Satisfaction With Life Scale [Pavot1993], a five-item instrument with established psychometric properties including high internal consistency, temporal stability, and convergent validity with related constructs. The review by Pavot and Diener confirms the scale's validity across diverse populations and its sensitivity to changes in life circumstances, making it appropriate for modelling as a continuous outcome variable. In the Canadian Survey dataset used in this thesis, life satisfaction is measured as a single-item numeric rating on a 0–10 scale, which is consistent with established single-item life satisfaction measures used in large-scale population surveys.

At the societal level, the World Happiness Report provides the most comprehensive annual assessment of national happiness across countries [Helliwell2023]. The primary measure is the Cantril Self-Anchoring Striving Scale - referred to as the Life Ladder - in which respondents rate their current life on a scale from 0 (worst possible life) to 10 (best possible life). The WHR framework identifies six explanatory factors that account for variation in national happiness scores: GDP per capita, social support, healthy life expectancy, freedom to make life choices, generosity, and perceptions of corruption. These factors are precisely the predictors available in the Gallup dataset analysed in this thesis, enabling a direct empirical test of whether their importance rankings recovered by machine learning and Bayesian Network models are consistent with the established WHR framework.

The OECD [OECD2020] provides a complementary framework for measuring societal wellbeing that extends beyond happiness to include material conditions, quality of life, sustainability, and inequality. The OECD guidelines distinguish between objective indicators - income, employment, housing - and subjective indicators - life satisfaction, affect balance, meaning - and emphasise the importance of measuring both dimensions for a complete picture of population wellbeing. This distinction motivates the inclusion of both objective socioeconomic variables (income, food security, employment) and subjective psychological states (mental health, stress, sense of belonging) as predictors in the Canadian Survey models.

Chapter 3 – Methodology

3.1. Overview

This chapter describes the methodological framework used to address the five research questions set out in Chapter 1. The study adopts a comparative design in which four families of models - statistical regression, machine learning, hybrid, and Bayesian Networks - are applied to five datasets varying in scale, measurement level, and task type. Two datasets involve classification of psychological outcomes at the individual level; one involves regression of life satisfaction at the individual level; and one involves regression of subjective wellbeing at the societal (macro) level. A fifth dataset, synthetically generated, is retained as a data quality case study rather than a substantive modelling exercise.

Section 3.2 describes each dataset and the preprocessing decisions applied to it. Section 3.3 introduces the four model families and their configurations. Section 3.4 covers the evaluation framework, including metrics, cross-validation strategy, and the handling of the evaluation asymmetry between ML and BN pipelines. Section 3.5 gives a full account of the Bayesian Network structure learning approach, covering the algorithm, scoring function, expert knowledge constraints, parameter estimation, and inference procedure.

3.2. Dataset descriptions and preprocessing decisions

Five datasets were selected to cover a range of wellbeing constructs, measurement instruments, and analytical challenges. The distinction between micro-level (individual survey data) and macro-level (country-aggregated data) is treated as a primary axis of comparison throughout the thesis.

3.2.1 Mental Health in Tech Survey

The OSMI Mental Health in Tech Survey is a publicly available dataset collected via self-report from technology-sector workers. After removing observations with missing target values, the working sample comprised 995 respondents and 43 features. The prediction target was a three-class variable, *combined_target*, constructed by crossing the work interference item (*work_interfere*) with whether the respondent had sought treatment: Class 0 indicates no interference, Class 1 indicates interference without treatment, and Class 2 indicates interference with treatment. The class distribution was approximately 60.6%, 21.4%, and 18.0% respectively.

Preprocessing for the machine learning pipeline included the addition of binary missingness indicators for features with non-trivial missing rates, followed by median imputation for continuous variables and mode imputation for categoricals. Nominal features were one-hot encoded with reference categories dropped to prevent multicollinearity. For the Bayesian Network pipeline, missing values were preserved as a distinct state rather than imputed, ordinal encodings were retained, and high-cardinality geographic features (*Country* and *state*) were excluded.

3.2.2 Student Depression Dataset

This dataset contains survey responses from students on academic, social, and psychological variables. After filtering to confirmed student respondents - removing 0.1% of rows identified as non-students - the sample comprised 27,870 observations. The binary target, *Depression_yes*, indicated presence or absence of depression. Four features were dropped prior to modelling: *City*, which contained a mix of city names and degree abbreviations indicating data entry inconsistency; *Profession*, redundant after student filtering; and *Work Pressure* and *Job Satisfaction*, which are not applicable to a student population. The final feature set comprised 23 variables.

A notable characteristic of this dataset is the dominance of the suicidal ideation feature, which accounts for approximately 52–56% of intrinsic variable importance in tree-based models. Because suicidal ideation is a co-symptom of depression rather than an antecedent cause, its concentration of predictive signal raises questions about whether the models are learning genuine risk-factor structure or simply detecting a correlated symptom. This is examined in sensitivity analysis. For the Bayesian Network, age and CGPA were discretised into bins to satisfy the discrete variable requirement; continuous features in the ML pipeline were retained in their original form.

3.2.3 Canadian Community Health Survey

This large-scale national health survey comprises approximately 108,000 respondents. The continuous target, *Life_satisfaction*, is measured on a 0–10 scale. Columns with more than 60% missing values were dropped prior to modelling. After preprocessing, approximately 58 features were retained.

Two variables - *Mental_health_state* and *Gen_health_state* - are self-rated health variables that function as proxies for life satisfaction: they correlate strongly with the target but arguably reflect a similar evaluative process rather than being independent causal inputs. Their contribution is examined in a dedicated sensitivity analysis.

3.2.4 WHR Synthetic Dataset

This dataset, presented on Kaggle as a World Happiness Report synthetic dataset, contains 4,000 observations across 10 countries and 20 years. The *Happiness_Score* column was dropped prior to modelling due to direct target leakage. Initial modelling revealed that all model families returned R^2 values near zero, and the maximum absolute correlation between any feature and the target was 0.049. It is retained as a case study in data quality detection, illustrating how a consistent modelling pipeline can identify a null-signal dataset before interpretability or causal analysis is attempted. The methodological implications of this finding are discussed in Section 4.4.1.

3.2.5 WHR Gallup Dataset

The Gallup World Poll data underlying the World Happiness Report was obtained via Kaggle and represents the most credible macro-level dataset in this study. The working sample comprised 1,949 observations across 166 countries and the period 2005–2020. The target variable, *Life_Ladder*, is the Cantril self-anchoring scale, which asks respondents to place their current life on a ladder from 0 (worst possible life) to 10 (best possible life). Seventeen features were retained after preprocessing, including log GDP per capita, social support, healthy life expectancy at birth, freedom to make life choices, generosity, perceptions of corruption, positive affect, negative affect, year, and binary missingness indicators for all columns with missing values. The maximum absolute feature-target correlation was 0.783, confirming genuine predictive signal. For the Bayesian Network, all continuous variables were quantile-discretised into five equal-frequency bins; year was binned into five-year periods to reduce the number of discrete states while preserving temporal ordering.

3.3. Model families and configurations

Four model families are applied consistently across all datasets. Each family represents a distinct position on the trade-off between predictive performance and interpretability, and their comparison is central to Research Questions 1, 2, and 3.

3.3.1 Statistical Baseline Models

The statistical baseline models serve as the interpretability anchor for the study. For classification tasks, logistic regression is used; for regression tasks, ordinary least squares linear regression is used. These models are selected not because they are expected to achieve the highest predictive accuracy, but because they are maximally transparent: every predictor's effect is directly readable from its estimated coefficient, including direction and magnitude. This transparency makes them the natural reference point for the interpretability comparisons developed in later chapters.

A single decision tree (classifier or regressor depending on task) is also included in the baseline set. Decision trees occupy an intermediate position: they are non-parametric and capable of capturing non-linear relationships and interactions, yet their structure can be visualised and traversed directly. Maximum depth is constrained via cross-validation to limit overfitting.

3.3.2 Machine Learning Models

Two ensemble methods form the primary machine learning models. Random Forest constructs a large ensemble of decision trees, each trained on a bootstrap sample of the training data with a random subset of features considered at each split. Predictions are aggregated by majority vote (classification) or averaging (regression). The randomisation introduced at both the sample and feature level reduces variance relative to any individual tree, and the ensemble's feature importance scores - based on the aggregate reduction in node impurity across all trees - provide a global measure of variable relevance.

XGBoost implements gradient boosting, constructing trees sequentially such that each new tree is fitted to the pseudo-residuals of the current ensemble under a differentiable loss function. Regularisation is applied through a learning rate (shrinkage), subsampling of rows and columns, and L1/L2 penalties on leaf weights. XGBoost achieves the highest or near-highest predictive performance across all datasets in this study. Its predictions are used as the basis for the SHAP interpretability analysis described in Section 3.4.

Both models are evaluated using five-fold cross-validation with stratification on the target variable. Hyperparameters are held at well-established defaults throughout; the primary goal is comparison across model families rather than maximum optimisation of any individual model.

3.3.3 Hybrid Models

Hybrid models combine the feature selection capacity of ensemble methods with the interpretability of linear models. Two selection-then-estimation pipelines are evaluated: one using Random Forest importance scores to select features, followed by logistic or linear regression on the reduced feature set; and one using XGBoost importance scores in the same role. A third hybrid uses stacking, training a logistic or linear regression meta-learner on the out-of-fold predictions generated by Random Forest and XGBoost as base learners.

The inclusion of hybrid models reflects a practically motivated question: whether the interpretability of a linear model can be recovered without substantially sacrificing the predictive power of ensemble methods.

3.4. Evaluation framework

Evaluation metrics are chosen to match the task type and to reflect the priorities established in the research questions.

For classification tasks - MH Tech and Student Depression - the primary metric is macro-averaged F1 score, which computes precision and recall separately for each class before averaging, weighting all classes equally regardless of their frequency in the data. This is appropriate given the class imbalance present in both datasets, where a majority-class classifier would score misleadingly well on accuracy. Area under the ROC curve (AUC) is reported as a threshold-independent complement; for the three-class MH Tech target, AUC is computed using a one-vs-rest strategy.

For regression tasks - Canadian Survey, WHR Gallup, and WHR Synthetic - the primary metric is R^2, the proportion of variance in the target explained by the model. Root mean squared error (RMSE) is reported alongside R^2 to provide a metric expressed in the original scale of the target variable, which aids interpretability of effect magnitudes.

All machine learning models are evaluated using stratified five-fold cross-validation, and the reported metrics are the mean values across the five held-out folds. Cross-validated metrics are the primary basis for performance comparisons, as they provide an estimate of out-of-sample generalisation rather than in-sample fit. Bayesian Network predictions are computed on the full dataset because the pgmpy inference pipeline does not readily support fold-level re-estimation of structure and parameters. This evaluation asymmetry - CV metrics for ML models versus full-data metrics for BNs - is acknowledged throughout the results and discussion chapters.

3.5. BN structure learning approach

Bayesian Networks are probabilistic graphical models that represent the joint distribution of a set of random variables as a directed acyclic graph (DAG). Each node in the graph represents a variable; each directed edge from node A to node B encodes a direct conditional dependence of B on A, with B's distribution specified as a conditional probability table (CPT) over the states of its parents. The full joint distribution factorises as the product of these local conditional distributions, which makes inference computationally tractable and the learned structure directly interpretable as a set of conditional independence claims.

All Bayesian Networks in this study are implemented using pgmpy version 1.1.0.

3.5.1 Structure Learning Algorithm

Structure learning - the task of identifying the DAG from data - is performed using the Hill Climb Search (HCS) algorithm. HCS begins with an empty graph and iteratively proposes single-edge operations: addition, removal, or reversal of a directed edge. At each step, the operation yielding the greatest improvement in a scoring function is accepted; the algorithm terminates when no single-edge operation improves the score. HCS is a score-based greedy search and does not guarantee a globally optimal solution, but it is computationally practical for the variable counts encountered in this study and has been shown to recover plausible structures in comparable applied settings [Scutari2014].

To prevent overfitting through excessive parent sets, the maximum in-degree - the number of parents permitted for any single node - is constrained to three. This limit reduces the combinatorial search space and produces structures that are more readily interpretable.

3.5.2 Scoring Function

The scoring function used throughout is BIC-d, the Bayesian Information Criterion adapted for discrete data. For a DAG G and dataset D of n observations, the BIC score is:

$$\mathrm{BIC}(G, D) = \log P(D \mid G, \hat{\theta}) - (k / 2) \cdot \log n$$

where $\log P(D \mid G, \hat{\theta})$ is the log-likelihood of the data under the maximum likelihood parameter estimates $\hat{\theta}$, and k is the total number of free parameters across all CPTs in the graph. The penalty term $(k / 2) \cdot \log n$ discourages the addition of edges whose likelihood gain does not outweigh the cost of the additional parameters they introduce. HCS maximises this score, meaning structures with more edges are only preferred when they provide a sufficiently better fit to justify their complexity.

The pgmpy 1.1.0 built-in BIC scorer is incompatible with the BDeu parameter estimates used in this pipeline due to internal inconsistencies in how the library computes sufficient statistics for mixed prior and ML estimation. BIC scores are therefore computed using a custom implementation that directly evaluates the log-likelihood of the data under the learned CPTs and subtracts the standard BIC penalty. This implementation was validated against manually computed scores on small example graphs prior to use.

3.5.3 Expert Knowledge Constraints

An unconstrained Hill Climb Search will sometimes recover edges that are statistically supported by the data but domain-implausible - for example, an outcome variable appearing as a parent of a demographic variable, or a time-invariant characteristic being caused by a time-varying one. To quantify this problem, an unconstrained BN is estimated for each dataset and all recovered edges are evaluated against domain knowledge. Implausible edges are catalogued and the implausibility rate (number of implausible edges as a proportion of total edges) is reported.

A restricted BN is then estimated by encoding domain constraints as a forbidden edges list, passed to the HCS algorithm via pgmpy's ExpertKnowledge interface. Forbidden edges prevent the algorithm from ever placing a directed edge in a specified direction between a specified pair of nodes during the search. The categories of forbidden edges applied in this study include: demographic or background variables as targets of outcome or symptom variables; time-invariant variables as targets of time-varying ones; and any edge whose direction would imply a causal pathway that is biologically, temporally, or theoretically impossible given the survey design.

For the Canadian Survey, where the proxy status of *Mental_health_state* and *Gen_health_state* is a concern, an additional strongly restricted BN is estimated with those variables removed entirely from the structure search, producing a three-way comparison: unconstrained, restricted, and strongly restricted.

The BIC cost of applying constraints is defined as the percentage change in BIC score between the unconstrained and restricted models:

BIC cost (%) = ((BIC_restricted − BIC_unconstrained) / |BIC_unconstrained|) × 100

A positive value indicates the restricted model achieves a better (higher) BIC score than the unconstrained model - a result that occurs when the unconstrained search has overfit to noise via implausible edges. A negative value indicates a fitness cost: the constraints force the algorithm away from statistically preferred structures. Reporting this cost across datasets provides a quantitative characterisation of how much the unconstrained algorithm's output conflicts with domain knowledge on each dataset.

3.5.4 Parameter Estimation

Conditional probability tables are estimated using the BayesianEstimator with a BDeu (Bayesian Dirichlet equivalent uniform) prior. BDeu assigns a uniform prior over the parameters of each CPT, distributed equally across all parent configurations, with the total prior weight controlled by the equivalent sample size hyperparameter. An equivalent sample size of 10 is used throughout, which provides mild smoothing of sparse cells without substantially distorting CPTs that are well-supported by data.

3.5.5 Inference and Prediction

Probabilistic inference is performed using Variable Elimination (VE). VE is an exact inference algorithm that computes the posterior distribution of a query variable given observed evidence by systematically eliminating non-query, non-evidence variables through marginalisation. For a query variable with evidence e, VE returns the full conditional distribution P(target | e), from which the most probable state (for classification) or the expected value computed from bin midpoints (for regression) is used as the point prediction.

For regression targets that have been quantile-discretised into bins, the expected value prediction is:

$\hat{E}[\text{target} \mid e] = \Sigma\_i \text{ midpoint\_i} \cdot P(\text{target} = \text{bin\_i} \mid e)$

where the sum is taken over all bins and midpoint_i is the midpoint of the original continuous range corresponding to that bin.

Chapter 4 - Results

4.1. Mental Health in Tech

4.1.1 Baseline ML Results

The four baseline models were evaluated on the three-class *combined_target* using five-fold stratified cross-validation. Results are reported in Table 4.1.

Table 4.1. MH Tech - Baseline ML cross-validated performance.

Model	CV Macro F1	CV AUC
Logistic Regression	0.457	0.685
Decision Tree	0.408	0.640
Random Forest	0.470	0.696
XGBoost	0.441	0.691

Random Forest achieved the highest macro F1 (0.470) and AUC (0.696) among the baseline models, with logistic regression and XGBoost performing comparably. The decision tree was the weakest baseline on both metrics, consistent with its tendency to overfit on noisy, moderate-sized samples without ensemble averaging.

The overall performance level across all baseline models is modest. Macro F1 values in the range 0.41-0.47 reflect the difficulty of the three-class prediction task, the class imbalance between the majority no-interference class (60.6%) and the two interference classes, and the inherent noise in self-report survey data. The relatively small gap between logistic regression and Random Forest - 0.013 F1 points - suggests that the predictive structure in this dataset is largely captured by linear relationships, with limited additional signal recoverable through non-linear modelling.

Figure 4.1 shows the confusion matrix for the Random Forest model. The model performs strongest on Class 2 (interference with treatment), correctly classifying 96 of 120 instances, reflecting the relatively concentrated signal in treatment-seeking behaviour. Performance is weakest on Class 1 (interference without treatment), with only 10 of 36 instances correctly classified and substantial confusion with both Class 0 and Class 2. This asymmetry suggests that the intermediate class - experiencing interference but not seeking treatment - is the most difficult to distinguish, likely because it shares characteristics with both the no-interference and treatment-seeking groups.

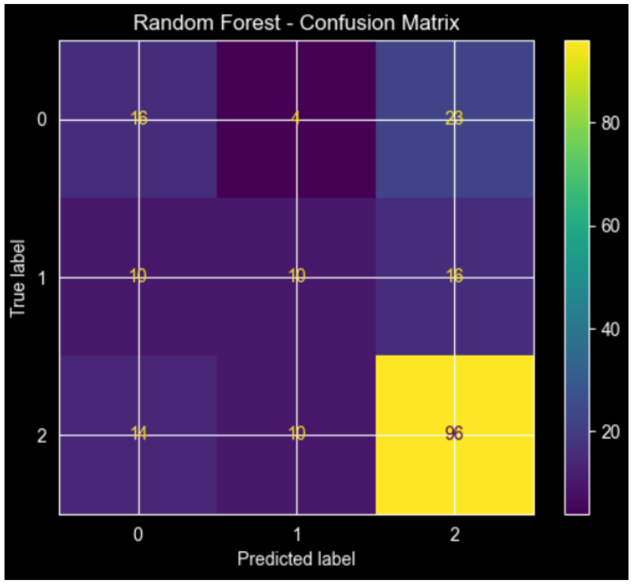


Figure 4.1 - Confusion Matrix: Best baseline model
(Random Forest) for MH Tech Dataset

Figure 4.2 shows the permutation feature importance for the Random Forest model. *family_history* is the dominant predictor by a substantial margin, with a permutation importance approximately three times larger than the next most important feature. *Gender_female*, *care_options_Yes*, and *seek_help_Don't know* follow at considerably lower importance. The dominance of *family_history* is consistent with the well-established heritability of mood and anxiety disorders and anticipates the BN finding in Section 4.1.3, where *family_history* is identified as a direct parent of *work_interfere*. The presence of employer-level features - *care_options*, *supervisor*, *benefits* - in the mid-importance range suggests that workplace environment contributes to the prediction beyond individual background factors, though their individual contributions are modest.

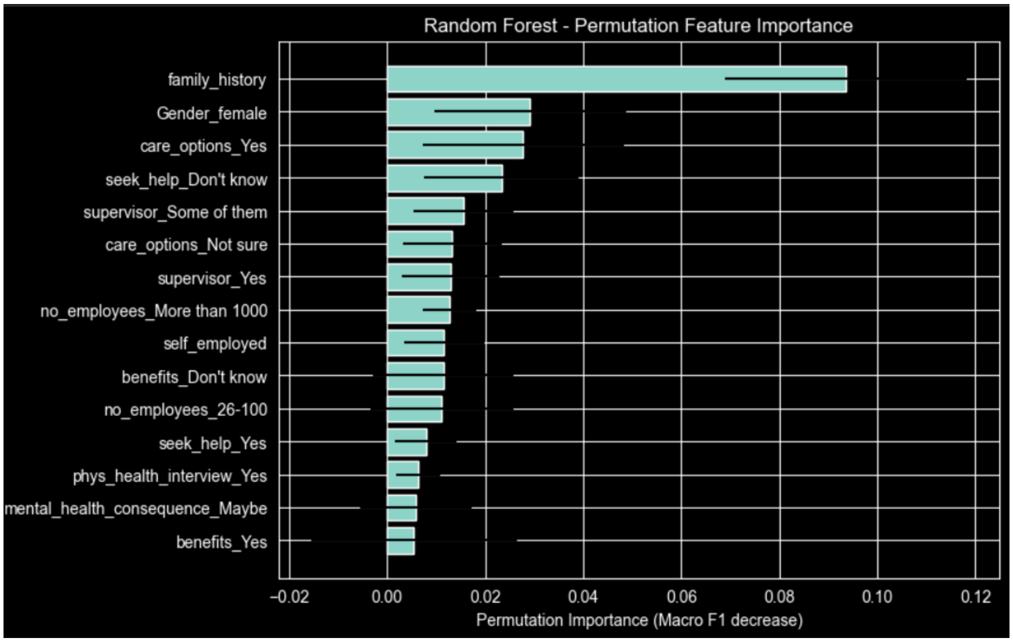


Figure 4.2 - Random Forest Permutation Feature
Importance for MH Tech Dataset

4.1.2 Hybrid ML Results

Three hybrid configurations were evaluated: Random Forest feature selection followed by logistic regression (RF → LogReg), XGBoost feature selection followed by logistic regression (XGB → LogReg), and stacking with a logistic regression meta-learner. Results are reported in Table 4.2.

Table 4.2. MH Tech - Hybrid ML cross-validated performance.

Model	CV Macro F1	CV AUC
RF Selection → LogReg	0.465	0.701
XGB Selection → LogReg	0.488	0.701
Stacking (LogReg meta)	0.460	0.700

The XGB → LogReg pipeline achieved the best hybrid F1 (0.488), a gain of 0.018 F1 points over the best baseline - the largest hybrid improvement observed across any dataset in this study yet still negligible in absolute terms. The cross-dataset pattern of hybrid model gains is discussed in Chapter 5.

4.1.3 BN Results

Two Bayesian Network versions were estimated: an unconstrained BN and a restricted BN incorporating domain-knowledge forbidden edges.

The unconstrained BN learned a graph containing 41 edges. Manual evaluation of these edges against domain knowledge identified 10 as implausible, yielding an implausibility rate of 24%.

The restricted BN was estimated after encoding a forbidden edges list to prevent the algorithm from recovering these implausible structures. The BIC cost of applying these constraints was −3.98%, indicating a modest fitness cost: the restricted model trades a small amount of statistical fit for a structurally coherent graph.

The restricted BN produced the following direct parents for the target nodes:

- Parents of *work_interfere*: Gender, family_history
- Parents of *treatment*: Gender, work_interfere

This structure encodes a two-stage pathway in which gender and family history of mental illness jointly influence the degree to which mental health affects work, and in which both gender and the experience of work interference influence whether an individual seeks treatment. The identification of *family_history* as a direct parent of *work_interfere* is consistent with the well-established heritability of mood disorders and anxiety, and with findings from the SHAP analysis described in Section 4.1.4.

Predictive performance of the restricted BN, evaluated on the full dataset, yielded a macro F1 of 0.471 and an AUC of 0.771. The F1 is comparable to the best baseline ML model (0.470), and the AUC of 0.771 exceeds the ML baselines (maximum 0.701), but given the evaluation asymmetry — full-data metrics for the BN versus cross-validated metrics for ML — both comparisons should be interpreted cautiously.

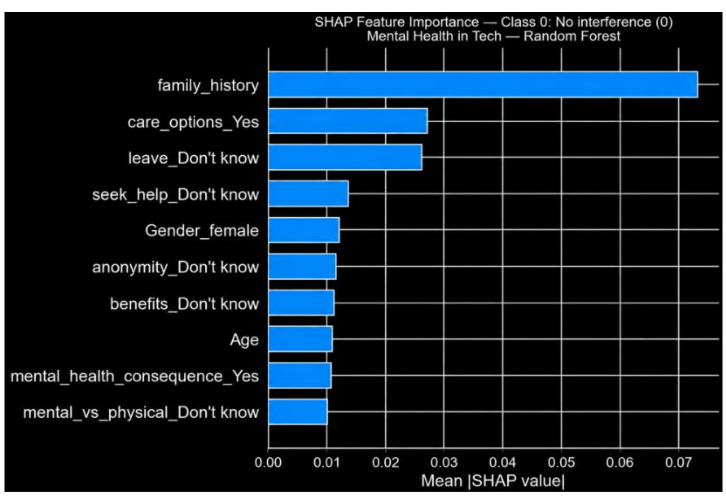
4.1.4 SHAP Analysis

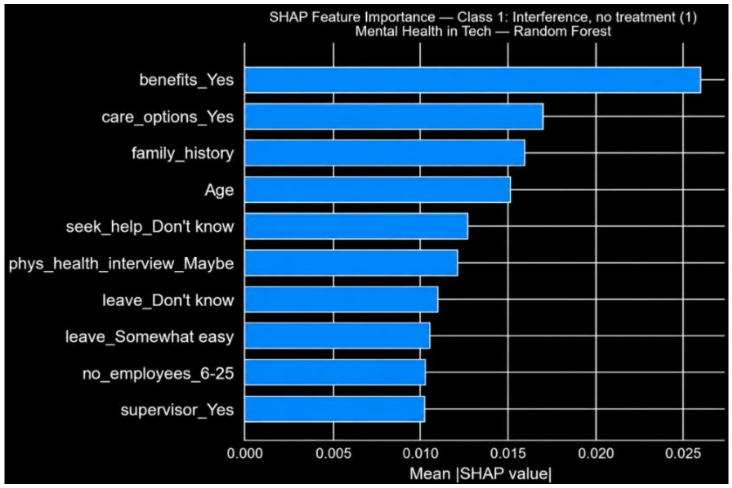
SHAP values were computed for the Random Forest model, which achieved the highest baseline F1. Because the target is three-class, SHAP values are computed separately for each class using a one-vs-rest decomposition, allowing the directional contribution of each feature to be examined per outcome.

The dominant predictor across all three classes is *family_history*. High values of *family_history* produce a strong positive push toward Class 2 (interference with treatment), and a corresponding negative push away from Class 0 (no interference). This pattern is consistent with the BN finding that *family_history* is a direct parent of *work_interfere* and suggests that familial risk is the single strongest marker of severe, treatment-seeking outcomes in this dataset.

care_options_Yes and *benefits_Yes* - variables indicating whether an employer offers mental health care options and benefits - are consistently among the top predictors across classes. Their directional effects indicate that awareness of available care options is positively associated with treatment-seeking (Class 2), which may reflect either genuine access effects or a selection effect in which more health-aware individuals are more likely both to seek treatment and to be aware of employer provisions.

Figures 4.4a, 4.4b, and 4.4c show the SHAP bar plots for Classes 0, 1, and 2 respectively. The contrast between classes is informative: *family_history* dominates Classes 0 and 2 but drops to third position for Class 1 (interference without treatment), where *benefits_Yes* and *care_options_Yes* become the leading predictors. This suggests that employer-level factors are most relevant for distinguishing the intermediate class - respondents experiencing interference but not yet seeking treatment - from the other two groups.





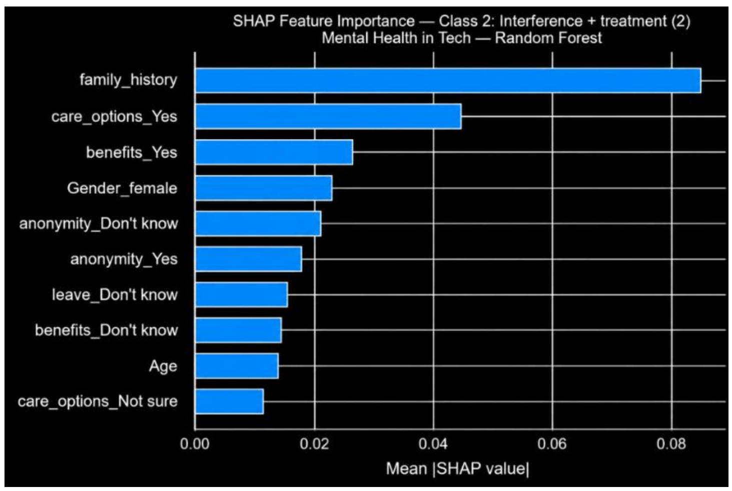


Figure 4.4 - SHAP Bar Plots: Class 0, Class 1, Class 2
for MH Tech Dataset

Figure 4.5 shows the beeswarm plot for Class 2. High values of *family_history* (pink) produce large positive SHAP contributions toward Class 2, while low values (blue) push strongly away from it - the clearest directional effect in the dataset. *care_options_Yes* and *benefits_Yes* show a similar pattern. *Gender_female* produces a modest positive push toward Class 2, consistent with research on gender differences in mental health help-seeking. The anonymity features show mixed directional effects, suggesting heterogeneous responses to workplace anonymity provisions across individuals.

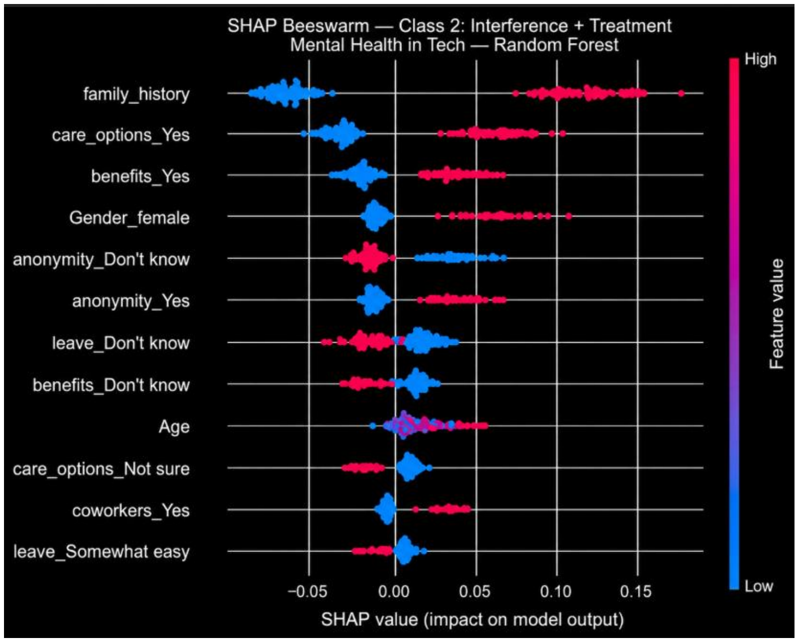


Figure 4.5 - SHAP Beeswarm: Class 2
(interference + treatment) for MH Tech Dataset

The overall SHAP profile for MH Tech is characterised by the dominance of a single background variable (*family_history*) over occupational and employer-level features, which has implications for the interpretation of workplace mental health interventions discussed in Chapter 6.

4.2. Student Depression

4.2.1 Baseline ML Results

The four baseline models were evaluated on the binary *Depression_yes* target using five-fold stratified cross-validation. Results are reported in Table 4.3.

Table 4.3. Student Depression - Baseline ML cross-validated performance.

TODO table

Model CV Macro F1 CV AUC

Logistic Regression 0.841 0.921

Decision Tree 0.822 0.901

Random Forest 0.837 0.917

XGBoost 0.841 0.921

Performance across all baseline models is substantially higher than for MH Tech, with macro F1 values ranging from 0.822 to 0.841. Logistic regression and XGBoost are tied for the highest F1 (0.841) and AUC (0.921). Random Forest performs only marginally below these, while the decision tree again underperforms relative to the ensemble methods.

The strong performance of logistic regression, matching the best ensemble model, indicates that the predictive structure in this dataset is well captured by linear decision boundaries. This is consistent with the concentration of signal in a small number of dominant features - in particular, suicidal ideation - whose effect on the target is large and approximately monotone.

4.2.2 Hybrid ML Results

Hybrid results are reported in Table 4.4.

Table 4.4. Student Depression - Hybrid ML cross-validated performance.

TODO table

Model	CV Macro F1	CV AUC
RF Selection → LogReg	0.839	0.921
XGB Selection → LogReg	0.840	0.921
Stacking (LogReg meta)	0.840	0.921

No hybrid configuration improves on the best baseline. The maximum hybrid F1 is 0.840, one thousandth below the XGBoost baseline of 0.841, and AUC is identical across all hybrid and baseline configurations at 0.921. This near-complete flatness of performance across model families is unusual and reflects the signal structure of the dataset: when a small number of features dominate prediction to the degree observed here, feature selection and meta-learning introduce negligible additional information.

4.2.3 BN Results

The unconstrained BN learned a graph containing 14 edges, of which 11 were evaluated as implausible against domain knowledge - an implausibility rate of 79%, the highest observed across all five datasets. The concentration of implausible edges in a dataset with a dominant co-symptom predictor is interpretable: the structure learning algorithm detects strong statistical associations between suicidal ideation, depression, and other variables, but encodes them in directions that reverse or misattribute causal priority. For example, depression appears as a parent of academic pressure in the unconstrained graph, a direction that is temporally and theoretically implausible given that academic pressure is an ongoing environmental stressor rather than a consequence of clinical diagnosis.

The restricted BN, estimated after applying the forbidden edges list, yielded a BIC cost of −0.29%, indicating that the domain constraints impose a negligible fitness cost while substantially improving the plausibility of the recovered structure.

The restricted BN identified the following direct parents of *Depression_yes*:

- Parents of *Depression_yes*: Academic Pressure, Financial Stress

This is a parsimonious and theoretically coherent structure. Academic pressure and financial stress are both well-established antecedent risk factors for depression in student populations (Helliwell et al., 2023; Diener, 1984), and their identification as direct parents - rather than suicidal ideation, which is a co-symptom - reflects the value of domain constraints in recovering substantively meaningful structure from data dominated by a proxy variable.

Predictive performance of the restricted BN on the full dataset was macro F1 = 0.826, AUC = 0.906. The gap between the BN and the best ML model (0.841 F1) is 0.015 F1 points, the smallest such gap across all datasets in this study. This near-parity suggests that the parsimonious BN structure - two direct parents - captures the majority of the predictive signal available in the dataset.

4.2.4 SHAP Analysis

SHAP values were computed for XGBoost, which matched logistic regression for the highest baseline F1 while providing richer non-linear structure for SHAP decomposition.

The most important feature by a substantial margin is suicidal ideation, with a mean absolute SHAP value of approximately 1.22. This dwarfs the second and third most important features: Academic Pressure (mean |SHAP| ≈ 0.95) and Financial Stress (mean |SHAP| ≈ 0.65). All directional effects are consistent with theory: high suicidal ideation, high academic pressure, and high financial stress all push predictions toward depression (positive SHAP contributions), while low values of each push predictions away from depression.

The dominance of suicidal ideation in the SHAP profile is important to interpret carefully. Suicidal ideation is not a cause of depression but a correlated symptom, and a model that relies primarily on this feature to achieve high F1 is not learning the kind of risk-factor structure that would be useful for early intervention or preventive design. The SHAP profile therefore reinforces the case for the sensitivity analysis in Section 4.2.5.

Beyond the top three features, CGPA (inversely associated with depression), dietary habits, and sleep duration contribute modest but consistent SHAP values in theoretically expected directions.

4.2.5 Sensitivity Analysis

To assess whether meaningful predictive structure remains once the dominant co-symptom proxy is removed, the full ML pipeline was re-run with suicidal ideation excluded from the feature set.

Table 4.5. Student Depression - Sensitivity analysis (without suicidal ideation).

TODO table

Model	CV Macro F1 (full)	CV Macro F1 (no suicidal ideation)	Δ F1
Logistic Regression	0.841	0.791	−0.050
XGBoost	0.841	0.791	−0.050
CV AUC (XGBoost)	0.921	0.873	−0.048

Removing suicidal ideation reduces macro F1 by 0.050 and AUC by 0.048 across all models. While this represents a meaningful reduction, the residual performance - F1 = 0.791, AUC = 0.873 - remains substantially above chance, confirming that genuine antecedent signal is present in the remaining features. After exclusion, the SHAP importance ranking shifts: Academic Pressure becomes the dominant predictor, followed by Financial Stress and CGPA, an ordering closely aligned with the BN's learned parent structure. This convergence between the BN structural result and the post-exclusion SHAP ranking provides mutually reinforcing evidence that these two stressors are the most substantively important modifiable risk factors in this dataset.

4.1. Canadian Survey

4.3.1 Baseline ML Results

The four baseline models were evaluated on the continuous *Life_satisfaction* target using five-fold cross-validation. Results are reported in Table 4.6.

Table 4.6. Canadian Survey - Baseline ML cross-validated performance.
TODO table

Model	CV R^2	CV RMSE
Linear Regression	0.406	1.300
Decision Tree	0.389	1.319
Random Forest	0.407	1.29
XGBoost	0.422	1.282

XGBoost achieved the best baseline performance (CV $R^2 = 0.422$, RMSE = 1.282), marginally outperforming Random Forest ($R^2 = 0.407$). Linear regression performed comparably to Random Forest, again indicating that a substantial proportion of the predictive structure is linear. The decision tree underperformed all other baselines.

An R^2 of 0.422 represents moderate predictive accuracy for a life satisfaction outcome measured at the individual level. Individual-level wellbeing is inherently subject to transient psychological states, measurement error, and idiosyncratic life circumstances that aggregate-level models do not face, and a ceiling on R^2 is therefore expected. The role of the self-rated health proxy variables in inflating this ceiling is examined in Section 4.3.5.

4.3.2 Hybrid ML Results

Results for the three hybrid configurations are reported in Table 4.7.

Table 4.7. Canadian Survey - Hybrid ML cross-validated performance.
TODO table

Model	CV R^2	CV RMSE
RF Selection → LinReg	0.404	1.303
XGB Selection → LinReg	0.406	1.301
Stacking (LinReg meta)	0.423	1.281

The stacking configuration marginally outperforms the best baseline ($R^2 = 0.423$ vs 0.422), a difference of 0.001 R^2 points. The selection-then-regression pipelines produce no improvement, with both falling slightly below the linear regression baseline. This pattern is consistent with the other datasets: feature selection followed by a linear model recovers most but not all of the variance captured by the linear baseline, and stacking provides at most a trivial incremental gain over the best single model.

4.3.3 BN Results

The unconstrained BN learned 85 edges, of which 28 were evaluated as implausible - an implausibility rate of 33%. This is the largest raw count of implausible edges across all datasets, reflecting the large number of variables (approximately 58) and the complex correlation structure of a broad national health survey.

Two restricted BN versions were estimated. The restricted BN applied the standard forbidden edges list encoding temporal, demographic, and causal implausibility constraints. The strongly restricted BN additionally excluded *Mental_health_state* and *Gen_health_state* from the structure search entirely, to assess whether their presence distorts the graph learned for the remaining variables.

The BIC cost of the restricted BN was +0.06% relative to the unconstrained model, indicating a marginal improvement in BIC score - meaning that the constraints removed statistically noisy edges as well as domain-implausible ones. The strongly restricted BN incurred a BIC cost of −4.43%, reflecting the meaningful fitness cost of excluding the two proxy variables that carry substantial shared variance with the target.

Both the restricted and strongly restricted BNs identified the same direct parents for *Life_satisfaction*:

- Parents of *Life_satisfaction*: Musculoskeletal_con, Food_security, Income_source

This structure is notable for several reasons. First, it is stable across both constrained versions, indicating robustness to the inclusion or exclusion of the proxy variables. Second, all three identified parents are modifiable social and economic determinants of health, consistent with the OECD framework for wellbeing measurement (OECD, 2020). Food security and income source represent material conditions of daily life; musculoskeletal conditions represent chronic physical health burden, which has a well-documented negative relationship with subjective wellbeing.

Predictive performance of the restricted BN on the full dataset was $R^2 = 0.076$, RMSE = 1.622; the strongly restricted BN yielded $R^2 = 0.078$, RMSE = 1.620. These values are substantially below the ML baselines (best CV $R^2 = 0.422$), representing the largest predictive gap between BN and ML observed in this study. This gap reflects the cost of constraining the BN to a parsimonious, causally plausible structure: a three-parent model cannot compete with an ensemble of hundreds of trees incorporating 58 features. The interpretive trade-off this represents - structural coherence at the cost of predictive coverage - is a central theme of the cross-dataset analysis in Chapter 5.

4.3.4 SHAP Analysis

SHAP values were computed for XGBoost, the best-performing baseline model.

The two dominant predictors are *Mental_health_state* (mean |SHAP| ≈ 0.40) and *Gen_health_state* (mean |SHAP| ≈ 0.31). Their combined SHAP contribution accounts for the majority of the model's predictive variance. The directional effects are asymmetric: poor mental and general health depress life satisfaction far more than good health elevates it, a pattern consistent with loss aversion effects documented in subjective wellbeing research (Diener, 1984). This asymmetry is visible in the beeswarm plots, where low health state values produce large negative SHAP contributions while high values cluster near zero.

Beyond the proxy variables, *Stress_level* is the third most important predictor, with higher stress consistently pushing life satisfaction downward. *Sense_belonging* appears as the fourth predictor with a positive directional effect - respondents reporting a stronger sense of community belonging report higher life satisfaction, consistent with social connection theories of wellbeing (Helliwell et al., 2023).

Food security and income source, identified by the BN as direct parents of the target, appear in the SHAP ranking but at lower importance than the proxy variables. This ordering reflects the statistical dominance of the proxies rather than their causal priority, a distinction the BN analysis is designed to address.

4.3.5 Sensitivity Analysis

To assess the contribution of the proxy variables and identify the predictors that remain important in their absence, the full ML pipeline was re-run after removing *Mental_health_state* and *Gen_health_state*.

Table 4.8. Canadian Survey - Sensitivity analysis (without proxy variables).

TODO table

Model	CV R^2 (full)	CV R^2 (no proxies)	ΔR^2
Linear Regression	0.406	0.298	−0.108
XGBoost	0.422	0.314	−0.108

Removing the proxy variables reduces CV R^2 by 0.108 across both models - the largest sensitivity-induced drop observed in this study. This confirms that the two self-rated health variables account for approximately a quarter of the total explained variance. The residual R^2 of 0.314 is non-trivial and indicates that genuine predictive structure remains in the remaining features.

After exclusion, *Stress_level* becomes the dominant predictor in the SHAP ranking, followed by *Sense_belonging* and *Food_security*. The emergence of food security as a top-three predictor in the proxy-excluded model is particularly notable: it converges with the BN's identification of food security as a direct parent of life satisfaction, providing cross-method evidence that material food access is a genuine upstream determinant rather than a correlate of the proxy variables.

4.1. World Happiness Report

4.4.1 Synthetic Dataset - Null Result and Methodological Discussion

The WHR synthetic dataset presented an anomalous result at the first stage of modelling: all model families returned R^2 values near zero, and no hybrid or BN configuration recovered any meaningful predictive structure. The maximum absolute correlation between any feature and the target *Life_Satisfaction* was 0.049, consistent with a dataset in which the target variable is effectively independent of all available predictors.

The Bayesian Network analysis confirmed this diagnosis: both the unconstrained and restricted BN structures learned zero edges, indicating that the scoring function found no feature-target association strong enough to justify even a single directed connection.

This outcome demonstrates a valuable property of the modelling pipeline adopted in this study: when applied to a null-signal dataset, it produces a coherent null result rather than an artefactually inflated one. A pipeline susceptible to overfitting would produce positive R^2 values on training data even in the absence of genuine signal; the cross-validated evaluation framework correctly returns near-zero performance estimates, and the BN's failure to learn any structure provides a complementary structural confirmation.

The methodological implication is that the WHR synthetic dataset cannot serve as a basis for claims about societal wellbeing predictors, and that its Kaggle presentation as a real-world dataset constitutes a data quality failure. The detection of this failure before any interpretability or causal analysis was attempted illustrates the importance of beginning comparative modelling studies with baseline signal verification. No further analysis of this dataset is reported.

4.4.2 Gallup Dataset - Results

4.4.2.1 Baseline ML Results

The four baseline models were evaluated on the continuous *Life_Ladder* target using five-fold cross-validation. Results are reported in Table 4.9.

Table 4.9. WHR Gallup - Baseline ML cross-validated performance.

TODO table

Model	CV R^2	CV RMSE
Linear Regression	0.763	0.543
Decision Tree	0.785	0.516
Random Forest	0.843	0.442
XGBoost	0.850	0.432

The WHR Gallup dataset yields the strongest predictive performance of any dataset in this study, with XGBoost achieving CV $R^2 = 0.850$ and RMSE = 0.432. Random Forest is close behind ($R^2 = 0.843$). The decision tree outperforms linear regression on this dataset - a reversal of the pattern seen in the micro-level datasets - suggesting that the relationship between societal indicators and life satisfaction contains meaningful non-linearities and interactions that a single tree is better positioned to capture than a global linear model.

The high R^2 values relative to the Canadian Survey (best baseline 0.422) reflect the fundamental difference between macro- and micro-level prediction. Country-level aggregates of GDP, social support, and institutional quality are structurally cleaner predictors of average national life satisfaction than individual-level survey responses are of any given person's satisfaction, as idiosyncratic noise is averaged away in the aggregation process.

4.4.2.2 Hybrid ML Results

Results for the hybrid configurations are reported in Table 4.10.

Table 4.10. WHR Gallup - Hybrid ML cross-validated performance.

TODO table

Model	CV R^2	CV RMSE
RF Selection → LinReg	0.759	0.547
XGB Selection → LinReg	0.755	0.551
Stacking (LinReg meta)	0.840	0.446

The hybrid models underperform the best baseline across all configurations. The selection-then-regression pipelines drop to $R^2 \approx 0.757$, a decline of approximately 0.09 R^2 points relative to the XGBoost baseline, indicating that forcing the non-linear ensemble's selected features into a linear model loses substantial predictive information on this dataset. Stacking recovers most of the gap ($R^2 = 0.840$) but still falls short of the XGBoost baseline (0.850). The WHR Gallup dataset is the only case in this study where hybrid models produce a performance decline rather than a neutral result, underscoring that the non-linear structure of macro-level societal data is not well served by linear meta-learners.

4.4.2.3 BN Results

The unconstrained BN learned 9 edges, of which 1 was evaluated as implausible - an implausibility rate of 11%, the lowest across all datasets. This result is notable: on the dataset with the highest data quality and strongest genuine signal, the unsupervised structure learning algorithm recovers a largely plausible graph without requiring domain correction. This suggests that the BN's tendency to learn implausible edges is amplified by noisy, high-dimensional survey data and attenuated when the underlying relationships are strong and consistent.

The restricted BN, estimated after applying the forbidden edges list to correct the single implausible edge, achieved a BIC cost of +8.77% - the best BIC improvement observed across all datasets. The positive sign indicates that the constraints actually improved the BIC score, meaning the single implausible edge in the unconstrained graph was statistically harmful as well as theoretically implausible, and its removal improved fit.

The restricted BN learned the following structure of direct relevance to the target:

- Parent of *Life_Ladder*: Log_GDP_per_capita (sole direct parent)
- Broader structure: Generosity → Health → GDP → Life_Ladder

This is a clean, interpretable causal chain. GDP per capita is the sole direct predictor of national life satisfaction in the BN's learned structure, with generosity and healthy life expectancy identified as upstream contributors to GDP. The structure is consistent with the economic and sociological literature on national wellbeing (Helliwell et al., 2023) and with the Easterlin tradition linking aggregate income to life satisfaction at the cross-national level.

Predictive performance of the restricted BN on the full dataset was $R^2 = 0.613$, RMSE = 0.880. The gap between the BN (0.613) and the best ML model (CV $R^2 = 0.850$) is 0.237 R^2 points - substantial, but smaller than the gap observed for the Canadian Survey in proportional terms. The BN achieves this performance using a single direct parent for the target, whereas XGBoost uses all 17 features with full non-linear interactions. The structural parsimony of the BN comes at a predictive cost, but the result demonstrates that a single-variable causal structure can account for more than 60% of the variance in national life satisfaction.

4.4.2.4 SHAP Analysis

SHAP values were computed for XGBoost, the best-performing model on this dataset.

The dominant predictor is *Log_GDP_per_capita*, with a mean absolute SHAP value of approximately 0.39. This is consistent with the BN's identification of GDP as the sole direct parent of *Life_Ladder* and provides convergent evidence from two independent analytical methods that GDP is the primary driver of cross-national variation in life satisfaction. All directional effects are consistent with the World Happiness Report theoretical framework: higher GDP, social support, life expectancy, freedom, and lower perceived corruption all push predictions toward higher life satisfaction. The effects are monotone and theoretically expected without exception, indicating that the XGBoost model has learned a coherent, interpretable representation of the macro-level wellbeing literature.

Social_support is the second most important SHAP predictor, followed by *Healthy_life_expectancy_at_birth* and *Freedom_to_make_life_choices*. Negative affect is a notable contributor in the negative direction, as expected. Generosity is among the weakest predictors by SHAP importance despite its appearance as an upstream node in the BN graph, consistent with the empirical literature suggesting that generosity's relationship to national life satisfaction is indirect and mediated by social capital effects.

4.4.2.5 Sensitivity Analysis

To assess whether positive and negative affect function as proxies for the target - effectively measuring similar evaluative states as life satisfaction itself - the full ML pipeline was re-run after removing both affect variables.

Table 4.11. WHR Gallup - Sensitivity analysis (without affect variables).

TODO table

Model	CV R^2 (full)	CV R^2 (no affect)	ΔR^2
Linear Regression	0.763	0.748	−0.015
XGBoost	0.850	0.832	−0.018

Removing positive and negative affect reduces CV R^2 by only 0.018 for XGBoost and 0.015 for linear regression. This is the smallest sensitivity-induced reduction in this study and indicates that the affect variables are not functioning as dominant proxies for the Cantril scale in the way that *Mental_health_state* and *Gen_health_state* do in the Canadian Survey. Their removal does not materially alter the SHAP importance ranking of the remaining variables: *Log_GDP_per_capita* remains dominant, and the ordering of social support, life expectancy, and freedom is preserved. This stability suggests that the WHR Gallup model's predictive structure is robust and not dependent on the inclusion of evaluatively similar affect measures.

Chapter 5 - Cross Dataset Analysis

5.1. Model comparison tables and figures

This section aggregates the cross-validated performance results from Chapter 4 into unified comparison tables, enabling direct assessment of how model families perform relative to one another across datasets and task types. Because the five datasets span two task types - classification and regression - results are presented in two separate tables.

5.1.1 Classification Datasets

Table 5.1 reports macro F1 and AUC for all model configurations across the two classification datasets. For Bayesian Networks, full-data metrics are reported and marked accordingly.

Table 5.1. Classification datasets - full model comparison (CV unless marked †).

TODO table

Model	MH Tech F1	MH Tech AUC	Student Depression F1	Student Depression AUC
Logistic Regression	0.457	0.685	0.841	0.921
Decision Tree	0.408	0.640	0.822	0.901
Random Forest	0.470	0.696	0.837	0.917
XGBoost	0.441	0.691	0.841	0.921
RF → LogReg	0.465	0.701	0.839	0.921
XGB → LogReg	0.488	0.701	0.840	0.921
Stacking	0.460	0.700	0.840	0.921
BN (restricted) †	0.471	0.771	0.826	0.906

Several patterns are immediately apparent. First, the performance gap between the two datasets is large and consistent across all model families. Student Depression achieves F1 values in the range 0.822–0.841 regardless of model complexity, while MH Tech achieves F1 values in the range 0.408–0.488. This gap does not narrow when more complex models are applied, indicating that it reflects differences in signal quality and task difficulty between datasets rather than the limits of any particular modelling approach. The Student Depression dataset benefits from a large sample (n = 27,870), a binary target, and a dominant predictor; MH Tech has a more balanced signal distribution, a three-class target, and a moderate sample size.

Second, the ordering of model families is inconsistent across the two datasets. On MH Tech, the hybrid XGB → LogReg pipeline achieves the highest F1 (0.488), marginally ahead of Random Forest (0.470). On Student Depression, logistic regression and XGBoost tie at the top (0.841) and no hybrid configuration improves on them. This inconsistency reinforces the general finding that no single model family consistently dominates across classification tasks.

Third, the BN achieves F1 near the ML baseline on both datasets - 0.471 versus the best CV F1 of 0.488 on MH Tech, and 0.826 versus 0.841 on Student Depression - despite using structurally constrained, parsimonious parent sets and being evaluated on the full dataset rather than cross-validated held-out folds. The BN AUC on MH Tech (0.771) appears to exceed all ML configurations (maximum 0.701), but this advantage is attributed to the evaluation asymmetry rather than structural superiority.

5.1.2 Regression Datasets

Table 5.2 reports R^2 and RMSE for all model configurations across the three regression datasets. WHR Synthetic is included for completeness; its near-zero values serve as a null baseline. BN metrics are full-data and marked accordingly.

Table 5.2. Regression datasets - full model comparison (CV unless marked †).

TODO table

Model	Canadian R^2	Canadian RMSE	Gallup R^2	Gallup RMSE	Synthetic R^2
Linear Regression	0.406	1.300	0.763	0.543	≈ 0.000
Decision Tree	0.389	1.319	0.785	0.516	≈ 0.000
Random Forest	0.407	1.299	0.843	0.442	≈ 0.000
XGBoost	0.422	1.282	0.850	0.432	≈ 0.000
RF → LinReg	0.404	1.303	0.759	0.547	≈ 0.000
XGB → LinReg	0.406	1.301	0.755	0.551	≈ 0.000
Stacking	0.423	1.281	0.840	0.446	≈ 0.000
BN (restricted) †	0.076	1.622	0.613	0.880	-

The most salient feature of Table 5.2 is the performance gap between the Canadian Survey and WHR Gallup across all model families. XGBoost achieves $R^2 = 0.850$ on Gallup versus 0.422 on the Canadian Survey - a difference of 0.428 R^2 points that persists across every model family tested. This gap is not an artefact of model choice and is interpreted in Section 5.3 in the context of the micro versus macro distinction.

The WHR Synthetic column confirms the null-signal diagnosis: every model family, including the BN, returns effectively zero performance. This unanimity across methods - from a simple linear model to a gradient boosted ensemble to a probabilistic graphical model - provides strong evidence that the dataset contains no learnable structure, and validates the cross-validation framework's ability to detect the absence of genuine signal.

The BN predictive gap is largest on the Canadian Survey (BN $R^2 = 0.076$ versus ML best of 0.422, a gap of 0.346) and smallest on the Student Depression dataset (BN F1 = 0.826 versus ML best of 0.841, a gap of 0.015). This variation in BN-versus-ML gap across datasets is explored further in Section 5.2.

5.1.3 Hybrid Model Gains

Table 5.3 summarises the performance gain of the best hybrid configuration over the best baseline ML model on each dataset, isolating the incremental contribution of hybridisation.

Table 5.3. Hybrid model gain over best baseline - all datasets.

TODO table

Dataset	Best Baseline	Best Hybrid	Metric	Gain
MH Tech	0.470 (RF)	0.488 (XGB→LR)	Macro F1	+0.018
Student Depression	0.841 (LR/XGB)	0.840 (Stacking)	Macro F1	−0.001
Canadian Survey	0.422 (XGB)	0.423 (Stacking)	CV R^2	+0.001
WHR Gallup	0.850 (XGB)	0.840 (Stacking)	CV R^2	−0.010

Hybrid gains are negligible across all four substantive datasets, ranging from −0.010 to +0.018. The maximum gain, observed on MH Tech, is 0.018 F1 points - a result that does not constitute a practically meaningful improvement given the measurement uncertainty in cross-validated estimates. On two of the four datasets, the best hybrid underperforms the best baseline. There is no dataset on which hybridisation delivers a clearly worthwhile return relative to simply selecting the best single model.

This finding has a practical implication: the additional complexity and reduced transparency introduced by selection-then-regression pipelines and stacking meta-learners does not purchase meaningfully better predictions in the wellbeing research context. If the goal is maximising predictive accuracy, XGBoost alone is the appropriate choice. If the goal is interpretability, logistic or linear regression alone provides comparable performance with full coefficient transparency. The hybrid occupies no stable niche between these poles.

5.2. BN consolidation

This section brings together the Bayesian Network results from Chapter 4 to assess what the BNs collectively reveal about the structure of wellbeing data, how the learned structures compare across datasets, and how the tension between statistical fit and domain plausibility varies with data characteristics.

5.2.1 Learned Causal Structures

Table 5.4 summarises the direct parents of the target variable identified by the restricted BN on each dataset, alongside the broader structural features of the learned graph.

Table 5.4. BN structural summary - all datasets.

Todo table

Dataset	Direct parents of target	Total edges	Implausible edges (unconstrained)	Implausibility rate	BIC cost of constraints
MH Tech	Gender, family_history	41	10	24%	−3.98%
Student Depression	Academic Pressure, Financial Stress	14	11	79%	−0.29%
Canadian Survey	Musculoskeletal_con, Food_security, Income_source	85	28	33%	+0.06%
WHR Gallup	Log_GDP_per_capita	9	1	11%	+8.77%

Four observations emerge from this consolidation.

First, the implausibility rate is inversely related to data quality and signal strength. WHR Gallup, with the strongest genuine signal (max $|r| = 0.783$), produces only 11% implausible edges in the unconstrained BN. Student Depression, dominated by a co-symptom proxy, produces 79%. MH Tech and the Canadian Survey fall between these extremes. This pattern suggests that when genuine causal signal is strong and consistent, the structure learning algorithm recovers plausible graphs without requiring domain correction; when signal is concentrated in a proxy or distributed noisily across many correlated variables, the algorithm exploits spurious associations that domain knowledge must override.

Second, the BIC cost of constraints is also related to data quality. For WHR Gallup, applying constraints improves BIC by +8.77%, meaning the forbidden edges list removes statistically harmful structure as well as implausible structure - the two properties coincide on the highest-quality dataset. For the Canadian Survey's restricted BN, the cost is near zero (+0.06%), indicating that the constraints are orthogonal to the statistical fit. For MH Tech (−3.98%) and the strongly restricted Canadian BN (−4.43%), there is a modest fitness cost. Only on Student Depression is the cost negligible (−0.29%), because the dominant implausible edges in the unconstrained graph carry minimal BIC weight relative to the signal concentrated in the suicidal ideation variable.

Third, the direct parent sets identified by the restricted BNs are parsimonious and theoretically coherent across all four datasets. In no case does the restricted BN identify a parent that is theoretically implausible or surprising given the domain literature. The structures converge with established theoretical frameworks: familial risk and gender as determinants of workplace mental health interference (MH Tech); academic and financial stress as antecedent risk factors for student depression; material conditions - food security and income - as determinants of individual life satisfaction; and economic development as the primary determinant of national life satisfaction. This coherence across four independent datasets, each with different measurement instruments, populations, and scales, provides cumulative evidence that BNs with appropriate domain constraints reliably recover substantively meaningful structure.

Fourth, the number of edges and complexity of the learned graph varies substantially across datasets, driven by the number of variables and the density of inter-feature correlations. The Canadian Survey's 85-edge unconstrained BN reflects the high dimensionality and multicollinearity of a national health survey; WHR Gallup's 9-edge unconstrained BN reflects the relatively small variable set and clean factor structure of the Cantril-scale data.

5.2.2 BN Predictive Performance Relative to ML

Table 5.5 summarises the predictive gap between the restricted BN and the best ML baseline on each dataset, along with the number of direct parents used by the BN.

Table 5.5. BN vs best ML baseline - predictive gap and structural parsimony.

TODO table

Dataset	Best ML (CV)	BN (full data)	Gap	BN direct parents
MH Tech	F1 = 0.488	F1 = 0.471	−0.0172	
Student Depression	F1 = 0.841	F1 = 0.826	−0.0152	
Canadian Survey	R^2 = 0.422	R^2 = 0.076	−0.3463	
WHR Gallup	R^2 = 0.850	R^2 = 0.613	−0.2371	

The predictive gap between BN and ML is smallest on the two classification datasets and largest on the two regression datasets. On the classification tasks, the BN achieves near-parity despite using only two direct parents for the target and being evaluated without cross-validation. On the regression tasks, the gap is substantial - particularly for the Canadian Survey - reflecting both the larger number of features available to the ML models and the inherent limitation of discretising a continuous target for BN inference.

The regression gap deserves careful interpretation. A BN predicting life satisfaction from three parents is not attempting the same task as an XGBoost model incorporating 58 features with full non-linear interactions; it is encoding a structural hypothesis about direct causes. The appropriate evaluative frame for the BN's regression performance is therefore not how closely it approaches the ML ceiling, but whether the structure it encodes is causally meaningful and whether the predictive shortfall is an acceptable cost for that structural interpretability. This trade-off is examined in the discussion of Research Question 3 in Chapter 6.

5.3. Micro vs macro comparison

A primary axis of comparison in this study is the distinction between micro-level datasets - individual survey responses - and the macro-level WHR Gallup dataset, which aggregates responses to the country-year level. This section examines how this distinction affects predictive performance, model behaviour, and the alignment between statistical and structural results.

5.3.1 Predictive Performance

The most striking quantitative difference between micro and macro datasets is the level of predictive performance achievable across all model families. Table 5.6 compares the best cross-validated R^2 achieved on the two regression datasets.

Table 5.6. Micro vs macro regression performance - best ML model.

TODO table

Dataset	Level	Best CV R^2	Best CV RMSE	Target scale
Canadian Survey	Micro	0.422	1.2820	–10 (individual)
WHR Gallup	Macro	0.850	0.4320	–10 (country mean)

The macro dataset achieves more than twice the R^2 of the micro dataset with the same model family (XGBoost) and an overlapping set of predictor concepts. Both datasets use a 0–10 life satisfaction scale; both include economic, health, and social predictors. The performance difference is therefore not attributable to target scale or predictor domain, but to the fundamental structural difference between individual-level and aggregate-level data.

At the individual level, life satisfaction is subject to measurement error in single survey items, transient mood states at the time of response, idiosyncratic life circumstances not captured by the available features, and genuine heterogeneity in how individuals interpret and use satisfaction scales. These sources of residual variance are irreducible given the available predictors, and they impose a ceiling on achievable R^2 that no model, however complex, can overcome. At the country level, these individual-level fluctuations average out across thousands of respondents, leaving a cleaner structural signal in which GDP, social institutions, and health infrastructure explain the systematic cross-national variation.

This has a practical implication for wellbeing research: models developed on macro-level data should not be expected to achieve comparable performance when applied to individual prediction, even when the predictor constructs appear similar. The apparent explanatory power of societal-level wellbeing models partly reflects the statistical benefits of aggregation rather than the identification of individually predictive factors.

5.3.2 Model Family Ordering

The ordering of model families by performance differs between the micro and macro contexts. On the Canadian Survey, the decision tree underperforms linear regression ($R^2 = 0.389$ vs 0.406), and the gap between linear regression and XGBoost is narrow (0.406 vs 0.422). On WHR Gallup, the decision tree outperforms linear regression (0.785 vs 0.763), and the gap between linear regression and XGBoost is large (0.763 vs 0.850).

This pattern suggests that non-linear relationships and feature interactions are more prevalent and more learnable in the macro-level data. The relationship between national GDP and life satisfaction is known to be log-linear with diminishing returns; the interaction between social support and institutional trust is a well-documented moderation effect in the cross-national literature. These structured non-linearities benefit tree-based models. At the individual level, the relationships between survey predictors and life satisfaction are noisier and more heterogeneous, offering less consistent non-linear structure for ensemble methods to exploit.

5.3.3 BN Structural Alignment with Theory

Both the micro and macro BNs identify theoretically coherent parent sets, but the alignment is tighter at the macro level. The WHR Gallup BN's structure - Generosity → Health → GDP →

Life_Ladder - maps cleanly onto the theoretical framework articulated in the World Happiness Reports (Helliwell et al., 2023), in which economic development mediates the relationship between social capital and subjective wellbeing. The BIC improvement achieved by applying constraints (+8.77%) further indicates that the domain knowledge and the statistical structure are mutually reinforcing on the macro dataset.

The micro-level BNs produce coherent but more modestly supported structures. The Canadian Survey BN identifies food security and income source as direct parents, which is theoretically sound but leaves the majority of variance unexplained. The MH Tech and Student Depression BNs identify plausible risk factors, but the high implausibility rates in their unconstrained versions (24% and 79% respectively) indicate that the statistical associations in individual survey data are substantially more ambiguous with respect to causal direction than those in country-level aggregates.

The macro-versus-micro distinction therefore cuts across both predictive performance and structural validity: aggregated societal data is not only easier to predict but easier to model causally, in the sense that its correlation structure more cleanly reflects the theoretical causal ordering documented in the literature.

5.4. SHAP cross-dataset patterns

Across the four substantive datasets, SHAP analysis reveals a set of recurring patterns that cut across measurement level, task type, and predictor domain. This section identifies these patterns and considers their implications for wellbeing research.

5.4.1 Proxy Dominance

In two of the four datasets, a single feature dominates the SHAP importance ranking by a margin large enough to substantially reshape the model's behaviour. In the Student Depression dataset, suicidal ideation accounts for the majority of predictive variance, dwarfing all other features. In the Canadian Survey, the combined contribution of *Mental_health_state* and *Gen_health_state* accounts for the majority of explained variance and reduces all other predictors to secondary status. In both cases, the dominant feature is not a genuine upstream cause of the outcome but an evaluatively similar proxy - a co-symptom or a parallel self-report of the same underlying state. This proxy dominance inflates headline performance metrics without providing the kind of actionable, antecedent risk factor knowledge that would inform intervention or policy. The sensitivity analyses confirm this interpretation: removing the proxies reduces performance by 0.050 F1 points (Student Depression) and 0.108 R^2 points (Canadian Survey) but does not collapse prediction, and the residual SHAP rankings converge on theoretically meaningful antecedent predictors - academic and financial stress in the former, and stress level and food security in the latter.

The implication is that SHAP importance rankings should not be interpreted at face value as rankings of causal or policy-relevant predictors. High SHAP importance is a property of statistical association, not causal priority, and when a dataset contains a strong proxy variable, SHAP will identify it as the most important predictor regardless of its theoretical status.

5.4.2 Convergence Between SHAP and BN Parents

A consistent pattern across datasets is that the variables identified by the restricted BN as direct parents of the target are also among the top predictors in the post-proxy-exclusion SHAP rankings, even though the two methods are methodologically independent. This convergence is observed in three of the four datasets:

In the Student Depression dataset, the BN identifies Academic Pressure and Financial Stress as direct parents; after removing suicidal ideation, these become the top two SHAP predictors. In the Canadian Survey, the BN identifies Food_security and Income_source as direct parents; after removing the proxy health variables, Food_security emerges as a top-three SHAP predictor. In WHR Gallup, the BN identifies Log_GDP_per_capita as the sole direct parent; it is also the dominant SHAP predictor both with and without affect variables included.

This convergence has a methodological implication: SHAP applied to a well-specified model from which proxy variables have been excluded and the BN applied with domain constraints tend to agree on what matters. The two methods approach the structure of wellbeing data from different directions - one through marginal contribution attribution, the other through conditional independence testing - and their agreement when both are carefully applied strengthens the evidential basis for the identified predictors. Where the two methods diverge - most notably on MH Tech, where SHAP highlights employer-level features that do not appear in the BN structure - the divergence itself is informative, pointing to associations that are statistically present but not structurally stable.

5.4.3 Directional Consistency

Across all four datasets, the directional effects identified by SHAP are consistent with theoretical expectations without exception. High family history of mental illness is associated with worse workplace mental health outcomes; high academic and financial stress are associated with depression; poor food security and lower income are associated with lower life satisfaction; higher GDP is associated with higher national life satisfaction. No SHAP analysis in this study produced a theoretically anomalous directional finding.

This directional consistency is reassuring but also unsurprising given that wellbeing research has a mature theoretical base and the predictors used in these datasets were drawn from established survey instruments. It suggests that the models are not learning artefactual or spurious associations from the data, and provides a baseline confidence that the predictive signals identified are at least directionally valid even where their causal status remains uncertain.

5.4.4 Asymmetric Effects

A recurring but less prominent pattern in the SHAP profiles is asymmetry between the positive and negative ends of predictor distributions. On the Canadian Survey, poor mental and general health depress life satisfaction substantially more than good health elevates it - a loss-aversion-type asymmetry consistent with negativity bias in subjective wellbeing (Diener, 1984). On WHR Gallup, the effect of GDP on life satisfaction shows diminishing returns at the upper end of the distribution, consistent with the well-documented log-linear income-happiness relationship.

These asymmetries are invisible to linear models and would be missed by coefficient-based interpretation of logistic or linear regression. Their detection is a specific contribution of the SHAP analysis, and they add nuance to the directional consistency noted above: while the direction of effects is uniformly as expected, the magnitude of effects is not symmetric, and policies or interventions targeting the negative tail of predictor distributions are likely to yield disproportionately larger wellbeing returns than equivalent improvements at the positive tail.

Chapter 6 - Discussion

6.1. Answers to each RQ

This section addresses each of the five research questions set out in Chapter 1 directly and systematically, drawing on the results reported in Chapters 4 and 5.

6.1.1 RQ1: How Do Different Modelling Approaches Perform When Predicting Wellbeing at Individual Versus Societal Levels?

The results provide a clear and consistent answer: predictive performance is substantially higher at the societal level than at the individual level, and this difference is robust across all model families tested. The best individual-level regression model (XGBoost on the Canadian Survey) achieves CV $R^2 = 0.422$, while the best societal-level model (XGBoost on WHR Gallup) achieves CV $R^2 = 0.850$. This gap of 0.428 R^2 points persists regardless of whether simple linear models, ensemble methods, or hybrid pipelines are used, indicating that it reflects a structural property of the data rather than a modelling artefact.

The explanation for this gap lies in the statistical consequences of aggregation. Individual survey responses are subject to transient mood effects at the time of response, measurement error in single-item scales, and idiosyncratic life circumstances that the available predictors do not capture. At the country level, these sources of noise average out across thousands of respondents, leaving a cleaner signal in which GDP, social institutions, and health infrastructure account for the systematic cross-national variation in mean life satisfaction. This implies that the explanatory frameworks developed in macro-level wellbeing research - such as those formalised in the World Happiness Reports - are strong predictors of average national outcomes but should not be expected to translate directly into individual-level prediction without substantial attenuation.

Among individual-level datasets, performance varies considerably depending on signal concentration. The Student Depression dataset achieves CV F1 = 0.841, the strongest individual-level result, but this performance is heavily driven by the presence of a co-symptom proxy. When that proxy is removed, performance drops to 0.791 - still above chance but more representative of what genuine antecedent predictors can achieve. The MH Tech dataset, without such a dominant proxy, yields a ceiling of 0.488 F1 across all model families, reflecting a more typical individual-level prediction problem in which signal is distributed across many features of modest predictive power.

The ranking of model families is also level-dependent. At the macro level, tree-based ensemble methods substantially outperform linear regression (R^2 gap of 0.087), and the decision tree itself outperforms linear regression, suggesting that structured non-linearities and interactions in societal data benefit from non-parametric approaches. At the individual level, the gap between linear regression and XGBoost is narrow (0.016 R^2 points on the Canadian Survey), and logistic regression matches XGBoost on Student Depression. This suggests that the predictive structure in individual survey data is predominantly linear, and the marginal benefit of ensemble complexity is small.

6.1.2 RQ2: How Does Interpretability Differ Between Regression, ML, and Bayesian Networks?

The three model families provide fundamentally different forms of interpretability, and these differences are consequential for how findings should be communicated and used.

Linear and logistic regression provide coefficient-based interpretability: every predictor has a single estimated effect size with a direction and magnitude that is directly readable from the model output. This interpretability is unconditional - it does not require post-hoc analysis tools and is available for any prediction. However, it is also limited: coefficients assume linear, additive effects and provide no information about non-linearities, interactions, or the heterogeneity of effects across subgroups. On the Canadian Survey and WHR Gallup, where non-linear effects are present (the asymmetry of health effects; the diminishing returns of GDP), coefficient-based interpretability misses structures that matter for policy.

Machine learning models, particularly XGBoost, achieve higher predictive accuracy but sacrifice direct coefficient interpretability. SHAP analysis partially recovers this interpretability by decomposing each prediction into additive feature contributions, providing both global importance rankings and local, instance-level explanations. The SHAP profiles in this study are directionally consistent with theory across all datasets, and the beeswarm plots reveal asymmetric effects invisible to linear models. However, SHAP interpretability is post-hoc and attribution-based rather than structural: it explains what the model has learned to associate with the outcome, not why those associations exist or what causal pathway they reflect. High SHAP importance does not imply causal relevance, as the proxy dominance findings demonstrate clearly.

Bayesian Networks provide a third form of interpretability that is qualitatively different from both coefficient tables and SHAP rankings: a directed graph encoding conditional independence relationships among all variables. The BN does not merely rank predictors by importance; it makes explicit claims about which variables directly influence which others, and in what direction. This structural interpretability is the most informative of the three for causal reasoning and for designing interventions, because it identifies not just what predicts the outcome but what the model posits as the direct causes of that outcome. The restricted BN parent sets in this study - family history and gender for workplace mental health interference; academic pressure and financial stress for student depression; food security and income for life satisfaction; GDP for national life satisfaction - represent structurally explicit causal hypotheses that coefficient tables and SHAP rankings do not provide.

The trade-off is that BN structural interpretability comes at a cost to predictive coverage. The BN models fewer direct parents of the target than the ML models use features, and the resulting predictions explain less variance. Whether this trade-off is worthwhile depends on the research goal: if the goal is maximum predictive accuracy, ML with SHAP is the appropriate tool; if the goal is identifying a parsimonious set of directly influential factors to guide intervention or policy, the BN structure provides information that neither coefficients nor SHAP can supply.

6.1.3 RQ3: Are Bayesian Networks Better at Capturing Plausible Causal Structures in Wellbeing Data?

The evidence from this study is nuanced and dataset-dependent. When applied to high-quality, moderate-dimensional data with strong genuine signal - as in WHR Gallup - the BN recovers a largely plausible structure even without domain constraints (implausibility rate of 11%), and the application of constraints improves both structural validity and BIC score simultaneously. In this context, the BN is well-suited to causal structure recovery, and the learned graph aligns closely with the theoretical framework established in the World Happiness Report literature.

When applied to individual-level survey data characterised by proxy variables, high dimensionality, or noisy correlation structures, the unconstrained BN performs poorly as a causal discovery tool. The 79% implausibility rate on the Student Depression dataset and the 33% rate on the Canadian Survey indicate that the structure learning algorithm, without domain guidance, frequently encodes statistically convenient but theoretically incoherent causal directions. This is not a failure specific to BNs: any unsupervised causal discovery method applied to observational survey data faces the fundamental problem that correlation does not uniquely identify causal direction, and that proxy variables produce associations indistinguishable from genuine causal pathways by statistical criteria alone.

The key finding is therefore that BNs are capable of capturing plausible causal structures, but this capacity is conditional on the application of expert knowledge constraints. The restricted BNs in this study consistently identify theoretically coherent parent sets, and the convergence of these parent sets with the post-proxy-exclusion SHAP rankings provides independent corroboration that the BN structures are not artefactual. Expert knowledge is not a limitation of the BN approach but an integral component of it: the BN provides a formal framework for incorporating domain knowledge as constraints and then learning the remaining structure from data, which is precisely what observational causal inference requires (Pearl, 2009; Morgan and Winship, 2015).

The comparison between unconstrained and restricted BN versions also produces a practically useful diagnostic: the BIC cost of constraints quantifies the tension between statistical fit and domain plausibility at the dataset level. A large BIC improvement under constraints (WHR Gallup: +8.77%) indicates that the data quality is high enough for statistical and domain criteria to reinforce one another. A large BIC cost under constraints (strongly restricted Canadian BN:

−4.43%) indicates that the domain-plausible structure is genuinely less statistically supported, and that interpretive caution about the causal claims is warranted.

6.1.4 RQ4: Can Bayesian Networks Reveal Causal Structures That ML Models Cannot?

Yes, with an important qualification about what is meant by revealing causal structure. Machine learning models, including XGBoost with SHAP, can identify which variables are statistically important and whether their effects are positive or negative, but they cannot distinguish between a variable that directly causes the outcome and a variable that is associated with the outcome through a common cause, a mediating pathway, or a proxy relationship. SHAP assigns high importance to whatever predicts the outcome most strongly, regardless of the nature of the underlying relationship.

The BN, by contrast, encodes a specific claim about the conditional independence structure of the joint distribution: the parents of a node are those variables that directly influence it, with all other associations explained by indirect pathways or shared causes. When the restricted BN identifies food security and income source as direct parents of life satisfaction in the Canadian Survey while placing mental health state and general health state elsewhere in the graph, it is making a structural claim that SHAP cannot make - namely, that material conditions have a direct influence on life satisfaction that is not fully mediated by health status. The SHAP analysis of the full model, dominated by the proxy health variables, cannot distinguish this structure because it is not designed to.

Similarly, the BN's identification of the Generosity → Health → GDP → Life_Ladder chain on WHR Gallup reveals a mediating pathway structure - generosity influences life satisfaction indirectly through its effects on health and economic development - that neither a coefficient table nor a SHAP ranking surfaces. The SHAP analysis ranks GDP as the most important predictor and generosity among the weakest, which is accurate in terms of direct statistical contribution but obscures the indirect pathway through which generosity operates. The BN makes this pathway explicit.

The qualification is that BNs reveal the conditional independence structure of the data, not ground-truth causal structure. On observational data, a DAG is identifiable only up to a Markov equivalence class - multiple graphs with different edge directions can encode the same conditional independences - and the Hill Climb Search algorithm makes choices within this equivalence class based on scoring rather than genuine causal identification. The structures identified in this study are therefore best interpreted as causally plausible hypotheses consistent with the data and with domain knowledge, rather than confirmed causal facts. This epistemic status is appropriate for observational wellbeing research and is explicitly acknowledged wherever BN structures are discussed.

6.1.5 RQ5: How Robust Are These Models to Missing Data, Noise, and Correlated Predictors?

Robustness varies systematically across model families and data characteristics.

Regarding missing data, the preprocessing strategy of adding binary missingness indicators alongside imputed values proved effective across all ML pipelines. The missingness indicators on the WHR Gallup dataset carried non-trivial SHAP importance, indicating that the pattern of missingness itself is informative - countries with missing data on certain variables are not randomly distributed. The BN pipeline's strategy of preserving missing values as a distinct state is theoretically cleaner but practically constrained by the sparsity it introduces into conditional probability tables for rarely-occurring missing patterns.

Regarding noise and weak signal, the WHR Synthetic case study demonstrates that the cross-validated ML pipeline correctly returns near-zero performance on a null-signal dataset, and the BN correctly learns zero edges. This robustness to null signal is a desirable property: a pipeline that overfits to noise would produce positive performance estimates and potentially misleading structural outputs. The consistent near-zero results across all methods provide confidence that positive results elsewhere in the study reflect genuine signal.

Regarding correlated predictors, multicollinearity is most acute in the Canadian Survey, where a large number of health and socioeconomic variables share substantial variance. Linear regression is most sensitive to multicollinearity, as it can produce unstable coefficient estimates when predictors are highly correlated. Ensemble methods are more robust because individual trees consider random feature subsets, effectively averaging over the correlated feature space. The BN handles correlated predictors through the conditional independence framework: if two features are correlated only because they share a common cause, the BN will encode that common cause rather than creating a direct edge between them. However, in practice the Hill Climb Search can struggle to distinguish genuine conditional independence from weak partial correlation in high-dimensional data, which contributes to the high edge counts and implausibility rates on the Canadian Survey.

The sensitivity analyses provide additional evidence on robustness to dominant features. On Student Depression, removing suicidal ideation reduces F1 by 0.050 but leaves a substantively meaningful residual model. On the Canadian Survey, removing the two proxy health variables reduces R^2 by 0.108 but the residual model remains predictive and coherent. On WHR Gallup, removing positive and negative affect reduces R^2 by only 0.018, confirming that the model's structure is robust to the exclusion of evaluatively similar variables. These sensitivity results collectively indicate that the models are not wholly dependent on any single predictor and retain meaningful structure when dominant features are removed.

6.2. Prediction-explanation trade-off

The prediction-explanation trade-off - the tension between models that predict well and models that explain clearly - is a central theme in the methodology and philosophy of statistical modelling (Breiman, 2001; Shmueli, 2010). The results of this study provide a concrete, multi-dataset illustration of how this trade-off manifests in applied wellbeing research.

The hybrid model findings speak most directly to the prediction side of the trade-off. The consistent failure of hybrid configurations to outperform the best single model - with gains ranging from −0.010 to +0.018 across all datasets - implies that there is no middle ground in which interpretability is recovered without predictive cost. The selection-then-regression pipelines reduce the feature set to those identified as important by ensemble methods, then fit a transparent linear model on that subset. In theory this should combine the feature relevance knowledge of the ensemble with the interpretability of the linear model. In practice it delivers neither the full predictive power of the ensemble nor the correct coefficient estimates of a properly specified linear model, because the feature selection step introduces instability and the linear model misspecifies the non-linear residual structure.

The explanation side is illustrated most clearly by the BN results. The Canadian Survey BN, constrained to a causally plausible three-parent structure, achieves $R^2 = 0.076$ against an ML baseline of 0.422. This is not a failure of the BN as a modelling tool; it is the cost of prioritising structural coherence over predictive coverage. A model that correctly identifies food security, income source, and musculoskeletal conditions as direct parents of life satisfaction is more useful for policy than a model that predicts life satisfaction with $R^2 = 0.422$ by exploiting 58 correlated features including proxy health variables, even if the latter appears superior by standard performance metrics.

Shmueli (2010) argues that the choice between predictive and explanatory modelling should be made explicitly and in advance, as the two goals require different model selection criteria, different validation strategies, and different interpretations of results. This study demonstrates that the choice also has different consequences for what is learned. The ML pipeline optimised for cross-validated R^2 identifies proxy-dominated feature rankings that are statistically accurate but causally ambiguous. The BN pipeline optimised for structural plausibility recovers parsimonious causal hypotheses that are less predictively powerful but more directly actionable.

The practical implication for wellbeing researchers is that the choice of modelling approach should be driven by the intended use of results. Predictive models are appropriate when the goal is screening, early warning, or classification of individuals or groups. Explanatory models - including BNs with domain constraints - are appropriate when the goal is identifying leverage points for intervention, understanding the mechanisms through which social determinants operate, or communicating findings to policy audiences who need to understand why an outcome occurs rather than simply how well it can be predicted.

6.3. Data quality findings

The WHR Synthetic null result and the proxy dominance findings on Student Depression and the Canadian Survey collectively constitute a set of data quality lessons that extend beyond the specific datasets involved.

The synthetic dataset case study demonstrates that data presented on public platforms under the label of real-world survey data can contain no genuine predictive signal. The unanimous null result across all model families - including the BN's failure to learn any edges - provides the most rigorous possible confirmation of this absence. Importantly, this diagnosis was available at the baseline modelling stage, before any interpretability or causal analysis was attempted. Researchers who proceeded directly to SHAP analysis or BN structure learning on this dataset would have been analysing noise. The lesson is that baseline signal verification - checking whether the target variable has any learnable relationship with the available predictors before investing in complex modelling - should be a standard first step in any comparative modelling pipeline applied to public datasets of uncertain provenance.

The proxy dominance findings raise a different data quality concern. The suicidal ideation variable in the Student Depression dataset is not a data error; it is a valid, reliably measured item. The problem is interpretive: a model that achieves high F1 by detecting suicidal ideation is not learning to predict depression onset from antecedent risk factors but is effectively detecting a correlated symptom. This is sufficient for some applications - if the goal is identifying individuals who may currently be depressed, detecting suicidal ideation is directly relevant. It is insufficient, and potentially misleading, for applications focused on early intervention, where the goal is identifying risk factors present before depressive episodes occur.

The Canadian Survey proxy variables - *Mental_health_state* and *Gen_health_state* - present a related but distinct issue. These are self-rated health items collected in the same survey wave as the life satisfaction target, and they likely reflect the same underlying evaluative state from different angles. A model that relies on these variables is essentially predicting life satisfaction from life satisfaction, which inflates R^2 without providing causal insight. The sensitivity analysis demonstrates that genuine predictive information remains after their removal, but the headline R^2 of 0.422 gives an overly optimistic impression of how much is understood about the determinants of individual life satisfaction.

These findings suggest that wellbeing datasets routinely contain features that are co-measurements or lagged reflections of the outcome rather than genuine upstream predictors, and that standard feature selection and importance metrics - including SHAP - will elevate these features to top positions without flagging their problematic status. Domain knowledge is necessary to identify proxy variables in advance of modelling, and sensitivity analysis is necessary to quantify their influence. Neither step is standard in applied machine learning pipelines, and both should be.

6.4. Limitations

Several limitations of this study should be acknowledged, organised by their relevance to specific analytical choices.

Observational data and causal inference. All five datasets are observational, and none supports causal identification without additional assumptions. The causal language used in discussing BN structures - parents, direct causes, pathways - is shorthand for conditional independence claims consistent with a causal interpretation under the assumption that the learned DAG approximates the true data-generating process. This assumption is untestable from the data alone. The DAGs identified here are better described as causally plausible hypotheses consistent with domain knowledge and the observed correlation structure, rather than confirmed causal facts. Genuine causal identification would require experimental variation, longitudinal data with temporal ordering of measurements, or instrumental variable designs, none of which are available in the datasets used.

Cross-sectional measurement. Four of the five datasets are cross-sectional, measuring all variables at a single point in time. This limits the BN's ability to establish temporal precedence, which is a necessary (though not sufficient) condition for causal inference. The forbidden edges lists encoding temporal constraints on the Canadian Survey and MH Tech BNs are based on theoretical temporal ordering rather than observed timing, and this ordering could be challenged for some variable pairs.

BN evaluation asymmetry. As noted throughout the results chapters, BN predictive performance is reported on the full dataset while ML performance is cross-validated. This asymmetry means that direct numerical comparisons between BN and ML metrics overstate BN performance relative to what would be observed in a true out-of-sample evaluation. The magnitude of this asymmetry is unknown but likely modest for the large datasets (Student Depression n = 27,870; Canadian Survey n ≈ 104,000) and potentially more consequential for MH Tech (n = 995). All performance comparisons in this study are qualified accordingly, and the primary interpretive weight is placed on structural findings rather than predictive metrics for the BN models.

Hyperparameter optimisation. ML models in this study are evaluated at well-established default hyperparameter settings rather than being systematically tuned via grid or random search. This choice is deliberate - the goal is comparison across model families rather than maximum performance of any individual model - but it means that the reported performance figures represent conservative estimates of what each model family can achieve with optimisation. The practical effect is likely to be largest for XGBoost, whose performance is most sensitive to learning rate and regularisation settings, and smallest for logistic and linear regression.

Dataset representativeness. The five datasets vary considerably in their representativeness of broader populations. The OSMI Mental Health in Tech survey is a convenience sample of technology workers who chose to complete a voluntary online survey, which introduces selfselection bias that limits generalisation to the broader working population. The Kaggle Student Depression dataset has an unclear sampling frame. The Canadian Community Health Survey is a nationally representative probability sample and is the most methodologically rigorous micro-level dataset used. The WHR Gallup data, while not a census, uses stratified random sampling within countries and is the most credible source for cross-national comparisons.

Discretisation for BN inference. The continuous variables in the regression datasets were quantile-discretised into five bins for BN estimation and inference. Discretisation inevitably loses information - the within-bin variation in the original variable is collapsed to a single representative value - and the choice of five bins is a pragmatic compromise between granularity and CPT sparsity. Different discretisation choices could produce different learned structures and different predictive performance estimates. The robustness of the key structural findings - in particular, the identification of consistent direct parent sets across the restricted and strongly restricted BN versions on the Canadian Survey - provides some reassurance that the main results are not highly sensitive to this choice, but a systematic sensitivity analysis over discretisation granularity was not conducted.

Generalisability of hybrid model findings. The finding that hybrid models provide negligible gains over the best baseline is specific to the datasets, model configurations, and feature set sizes used here. It is possible that hybrid approaches would show larger benefits in higher-dimensional settings where feature selection provides greater dimensionality reduction, or in datasets where the predictive relationship is partly linear and partly non-linear in a way that the selection-then-regression pipeline could exploit. The finding should therefore be interpreted as informative about the wellbeing research context rather than as a general claim about hybrid model utility.

Chapter 7 - Conclusion

7.1. Summary of findings

This thesis set out to compare statistical, machine learning, hybrid, and Bayesian Network models for predicting and explaining psychological wellbeing across five datasets spanning individual and societal levels of measurement. The comparative design, applied consistently across datasets varying in scale, task type, signal quality, and predictor domain, has produced a coherent set of findings that address each of the five research questions and contribute to the broader methodological literature on prediction versus explanation in wellbeing research.

The first and most robust finding is that predictive performance is substantially higher at the societal level than at the individual level, and this gap is independent of model choice. The best macro-level model achieves CV $R^2 = 0.850$ on the WHR Gallup dataset, while the best micro-level regression model achieves CV $R^2 = 0.422$ on the Canadian Survey. This difference reflects the statistical consequence of aggregation: country-level means suppress idiosyncratic noise and leave a cleaner structural signal in which GDP, social support, and institutional quality account for systematic cross-national variation. The implication is that the explanatory frameworks developed in the macro-level wellbeing literature have genuine predictive power at the societal level but cannot be expected to translate directly into individual-level prediction without attenuation.

The second finding is that hybrid models provide negligible predictive gains over the best single model across all datasets, with gains ranging from −0.010 to +0.018. This near-zero return on complexity holds consistently whether the task is classification or regression, whether the dataset is small or large, and whether the best baseline is a linear model or an ensemble. The practical implication is that the hybrid pipeline - designed to recover interpretability without sacrificing accuracy - delivers neither the full predictive power of the ensemble nor the clean interpretability of the linear model. Researchers seeking maximum predictive accuracy should use XGBoost directly; those seeking interpretable coefficients should use logistic or linear regression directly.

The third finding concerns the Bayesian Network results. When applied to high-quality data with strong genuine signal, as in the WHR Gallup dataset, the BN recovers a largely plausible causal structure without domain correction - only 11% of edges in the unconstrained graph are implausible - and the application of domain constraints simultaneously improves both structural coherence and BIC score. When applied to noisier individual-level survey data, the unconstrained BN frequently encodes implausible causal directions, with implausibility rates reaching 79% on the Student Depression dataset. In all cases, the application of domain knowledge constraints via forbidden edges lists produces structurally coherent and theoretically interpretable parent sets - family history and gender for workplace mental health interference; academic pressure and financial stress for student depression; food security, income source, and musculoskeletal conditions for individual life satisfaction; and GDP per capita for national life satisfaction - at a modest and quantifiable BIC cost.

The fourth finding is the convergence between BN structural results and the post-proxy-exclusion SHAP rankings. Across three of four datasets, the variables identified by the restricted BN as direct parents of the target emerge as the top SHAP predictors once dominant proxy variables are removed from the feature set. This convergence, achieved by two methodologically independent approaches, provides mutually reinforcing evidence that the identified predictors represent genuine signal rather than artefacts of either method. It also establishes a practical workflow: BN structure learning with domain constraints and SHAP analysis with proxy exclusion are complementary tools that together provide a more robust characterisation of wellbeing determinants than either method applied in isolation.

The fifth finding is methodological rather than substantive: the consistent application of a cross-validated pipeline across all datasets enables reliable detection of null signal. The WHR Synthetic dataset, presented on a public platform as real-world data, returned near-zero performance across every model family tested, and the BN learned zero edges. This unanimous null result, produced without any dataset-specific tuning or inspection, demonstrates that baseline signal verification is both possible and necessary as a first step in any comparative modelling study using public datasets of uncertain provenance.

7.2. Contributions

This thesis makes contributions at three levels: methodological, empirical, and practical.

Methodological contributions. The primary methodological contribution is the systematic application of a four-family comparative pipeline - statistical, machine learning, hybrid, and Bayesian Network - to wellbeing data at both individual and societal levels. Previous comparative studies in this domain have typically contrasted two model families on a single dataset; the five-dataset design adopted here enables generalisations about model behaviour that are not possible from single-dataset comparisons. The consistent finding that hybrid models add negligible value, that BN implausibility rates vary predictably with data quality, and that the macro-versus-micro performance gap is model-independent are all conclusions that require multi-dataset evidence.

A second methodological contribution is the formalisation of the BIC cost of constraints as a diagnostic metric. Reporting the percentage change in BIC score between unconstrained and domain-constrained BNs provides a quantitative characterisation of the tension between statistical fit and domain plausibility at the dataset level. This metric is dataset-agnostic and can be computed for any BN estimation exercise where domain knowledge is available, making it a transferable tool for assessing the reliability of BN structure learning in applied settings.

A third contribution is the demonstration that proxy variable identification and sensitivity analysis should be treated as standard components of the wellbeing modelling pipeline rather than optional additions. The proxy dominance findings on two of the four substantive datasets - where a single feature concentrates the majority of predictive variance without representing a genuine upstream cause - show that standard importance metrics, including SHAP, will systematically mislead researchers who do not check for this pattern. The sensitivity analysis protocol applied here, rerunning the full pipeline after removing candidate proxies and comparing the resulting importance rankings with BN parent sets, provides a replicable procedure for this check.

Empirical contributions. At the empirical level, this thesis provides a cross-validated, multi-method characterisation of the predictors of wellbeing across four distinct populations and measurement contexts. The consistent identification of food security and income source as direct parents of individual life satisfaction, independent of proxy health variables, contributes to the literature on material determinants of wellbeing and reinforces the OECD framework's emphasis on basic needs as foundational to subjective wellbeing measurement (OECD, 2020). The identification of academic pressure and financial stress as the primary modifiable antecedents of student depression - confirmed by both BN and post-proxy SHAP analysis - provides a more substantively useful risk factor characterisation than the headline model performance would suggest. The confirmation that GDP per capita is the sole direct parent of national life satisfaction in a BN estimated on Gallup data, with health and generosity operating as upstream mediators, aligns with and formally extends the structural claims of the World Happiness Report framework (Helliwell et al., 2023).

Practical contributions. The practical contribution of this thesis is a set of model selection guidelines for researchers and practitioners working in the wellbeing domain. These guidelines, grounded in empirical evidence rather than theoretical preference, are as follows: use cross-validated ensemble methods (XGBoost or Random Forest) when the goal is predictive accuracy; use logistic or linear regression when interpretable coefficients are required and the predictive penalty is acceptable; do not expect hybrid pipelines to deliver a meaningful improvement over the best single model; use BNs with expert knowledge constraints when the goal is identifying parsimonious causal structures, and interpret the BIC cost of constraints as a measure of how much the domain-plausible structure is statistically supported; always verify baseline signal quality before investing in interpretability analysis; and always conduct proxy variable sensitivity analysis before interpreting SHAP importance rankings as rankings of causal or policy-relevant predictors.

7.3. Future work

Several directions for future research follow naturally from the limitations and findings of this thesis.

Longitudinal data. The most significant limitation of the causal claims made in this thesis is the cross-sectional nature of four of the five datasets. Extending the BN pipeline to longitudinal survey data, where the temporal ordering of measurements can be directly incorporated into the forbidden edges structure, would substantially strengthen the evidential basis for the identified causal parent sets. Dynamic Bayesian Networks, which extend the static DAG framework to model transitions between time points, are a natural vehicle for this extension and would allow the analysis of how wellbeing trajectories over time are shaped by changing economic and social conditions.

Causal discovery beyond BNs. The BN approach adopted here is one member of a broader family of causal discovery methods. Score-based methods such as the Greedy Equivalence Search (GES) and constraint-based methods such as the PC algorithm make different assumptions about the data-generating process and have different sensitivity profiles with respect to sample size and latent confounding. A systematic comparison of these methods on the same wellbeing datasets would clarify whether the structural findings reported here are specific to the Hill Climb Search with BIC scoring or are robust across the causal discovery landscape.

Hyperparameter optimisation and AutoML. The deliberate choice to use default hyperparameter settings in this study means that the reported ML performance figures are conservative. A natural extension is to apply systematic hyperparameter tuning - via Bayesian optimisation or random search - to the ensemble methods on each dataset and assess whether the performance ceiling and the relative ranking of model families are altered. This would also allow a more precise quantification of the predictive cost of using interpretable models, as the gap between tuned XGBoost and logistic regression with optimised regularisation may differ from the gap observed at default settings.

Transfer learning and domain adaptation. A question left open by this thesis is whether the causal structures identified in one dataset generalise to other populations measuring the same constructs. The BN parent sets - academic pressure and financial stress for student depression; food security and income for life satisfaction - are theoretically general, but the specific conditional probability distributions learned from a Canadian national survey or a Kaggle student dataset may not transfer to other national contexts or educational systems. Cross-dataset transfer of BN structures, validated against held-out datasets from different countries or time periods, would provide empirical evidence on the generalisability of the identified causal hypotheses.

Intervention modelling. The BN framework naturally supports intervention modelling through Pearl's do-calculus: having learned a causal graph, one can query what the expected change in the target distribution would be if a variable were intervened upon - set to a specific value - rather than merely conditioned on (Pearl, 2009). This would allow the BN structures identified here to be used directly for policy simulation: for example, estimating the expected change in national life satisfaction if GDP per capita were increased by a given amount while holding other variables fixed, or estimating the reduction in student depression risk associated with a policy that reduces financial stress. Implementing this extension would move the analysis from causal description to causal prediction and would substantially increase the practical utility of the BN results.

Fairness and subgroup analysis. This thesis treats each dataset as a single population and does not examine whether the learned models perform equitably across demographic subgroups. In the MH Tech and Student Depression datasets, where gender, age, and other demographic variables are available, it would be valuable to assess whether predictive performance and the identified causal structures are consistent across subgroups or whether they mask heterogeneous relationships. Differential performance across groups would have implications for the deployment of any predictive model in a clinical or occupational health context, and differential causal structures would suggest that the single-population BN is a simplification of a more complex, group-stratified data-generating process.

Richer proxy detection methods. The proxy variable identification approach used in this thesis relies on domain knowledge combined with sensitivity analysis - a practical and transparent procedure but one that depends on the researcher's prior awareness of which variables are likely to be proxies. A more systematic approach would apply causal structure tests, such as tests for d-separation or the front-door criterion, to formally assess whether a variable's association with the target is fully mediated by other features in the graph. Integrating such tests into the standard modelling pipeline would reduce dependence on domain knowledge and make proxy detection more reproducible across research groups and application domains.

This thesis has demonstrated that the choice of modelling approach in wellbeing research is not merely a technical decision but a substantive one, with direct consequences for what is learned, what can be communicated, and what can be acted upon. The four model families examined here are not competing answers to the same question; they are tools suited to different questions, and their appropriate use depends on a clear articulation of whether the goal is prediction, explanation, or causal identification. The consistent application of all four families to the same datasets has made the costs and benefits of each choice empirically visible rather than theoretically assumed, and it is hoped that the guidelines and diagnostic tools developed here will be of use to researchers navigating this choice in their own work

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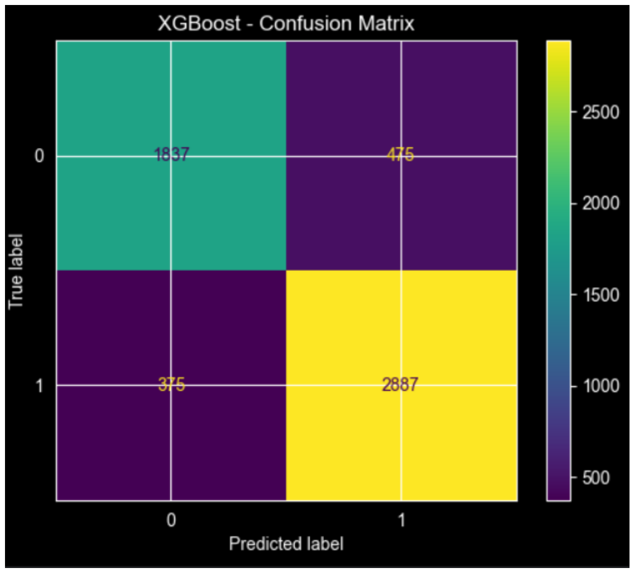

Figure 4.6 - Confusion Matrix: Best baseline model (XGBoost) for Student Depression Dataset

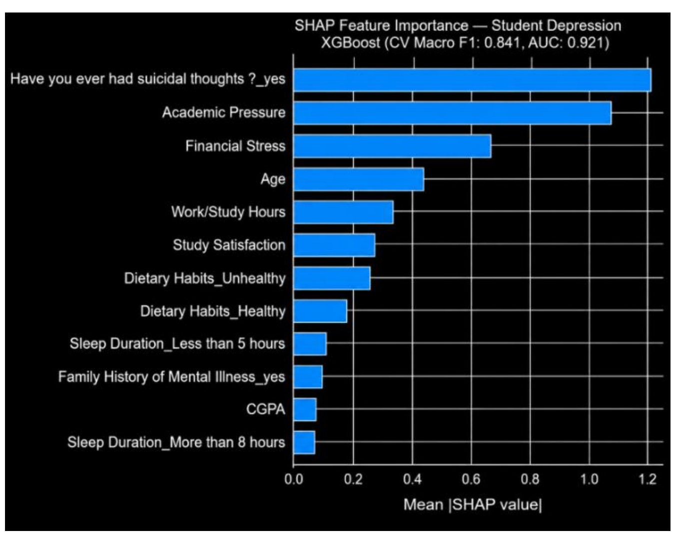

Figure 4.7 - SHAP Bar Plot: Full Model
for Student Depression Dataset

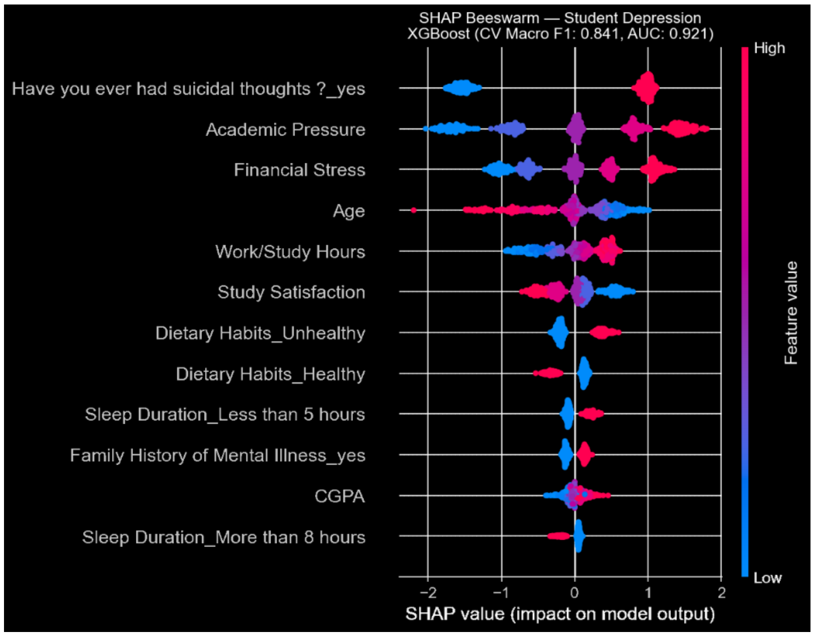

Figure 4.8 - SHAP Beeswarm: Full Model
for Student Depression Dataset

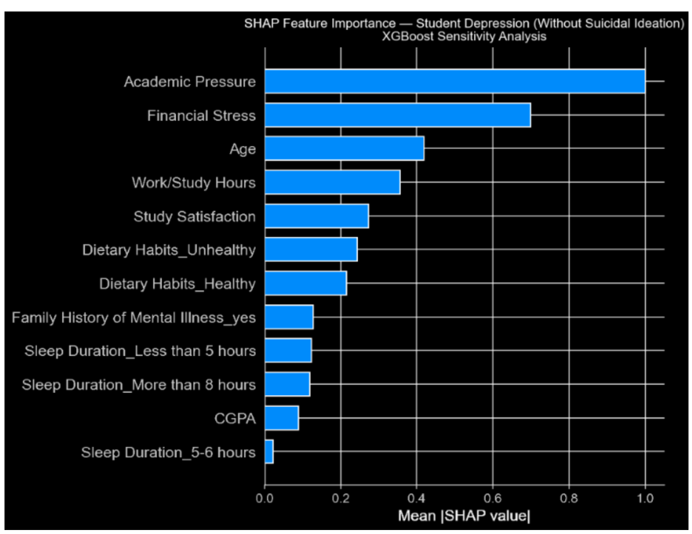

Figure 4.9 - SHAP Bar Plot: Sensitivity Analysis without
Suicidal Ideation for Student Depression Dataset

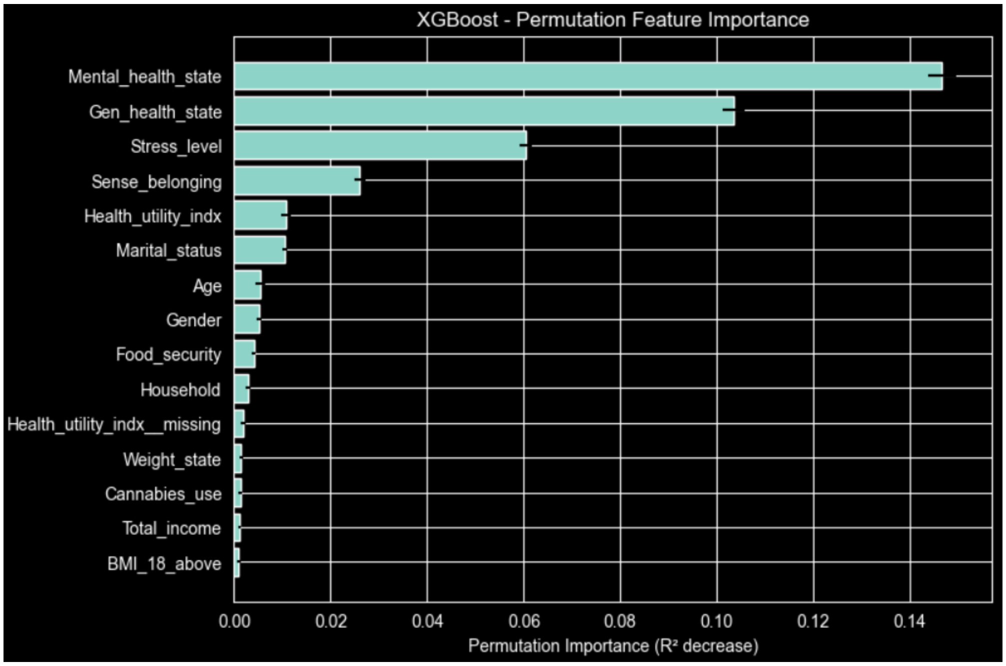

Figure 4.10 - XGBoost Permutation Feature Importance for Canadian Survey Dataset

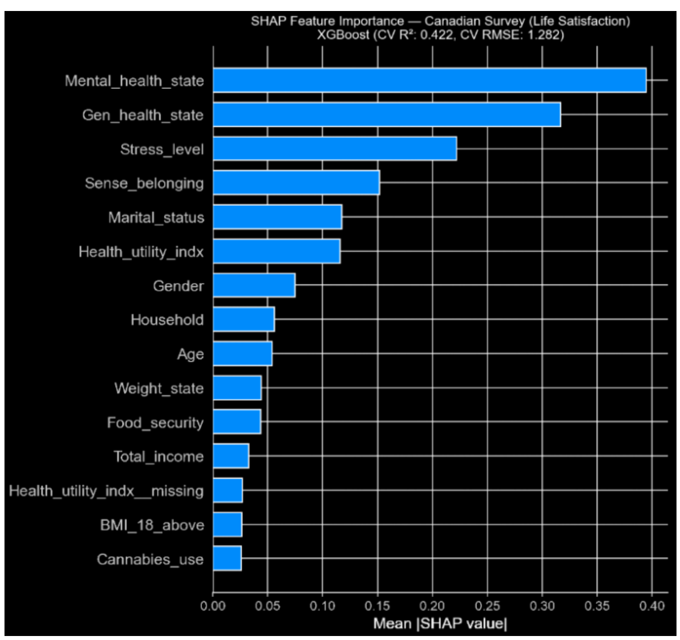

Figure 4.11 - SHAP Bar Plot: Full Model for Canadian Survey Dataset

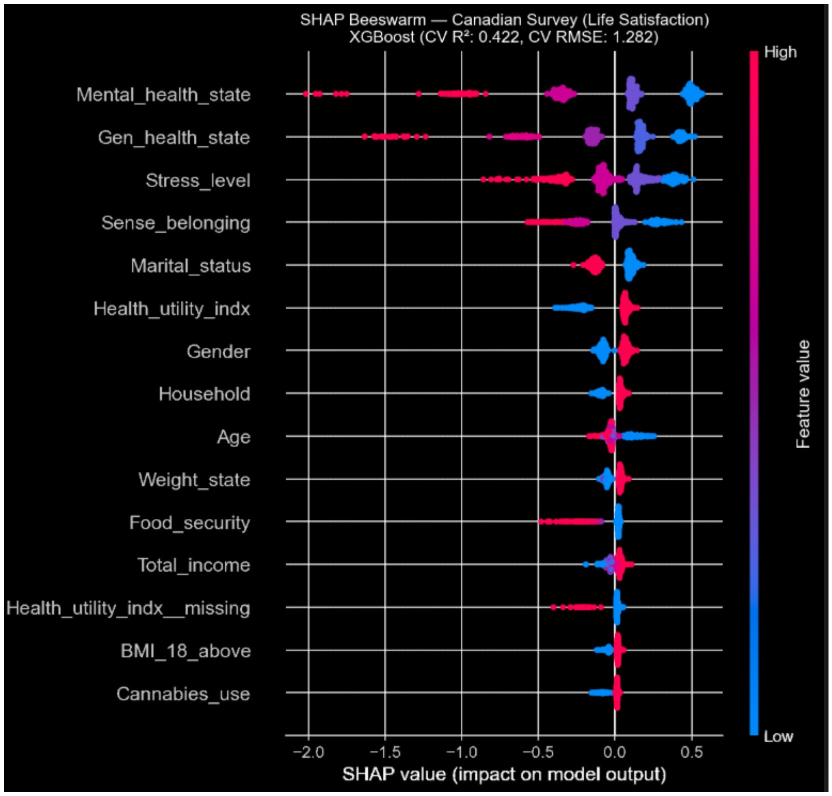

Figure 4.12 - SHAP Beeswarm: Full Model
for Canadian Survey Dataset

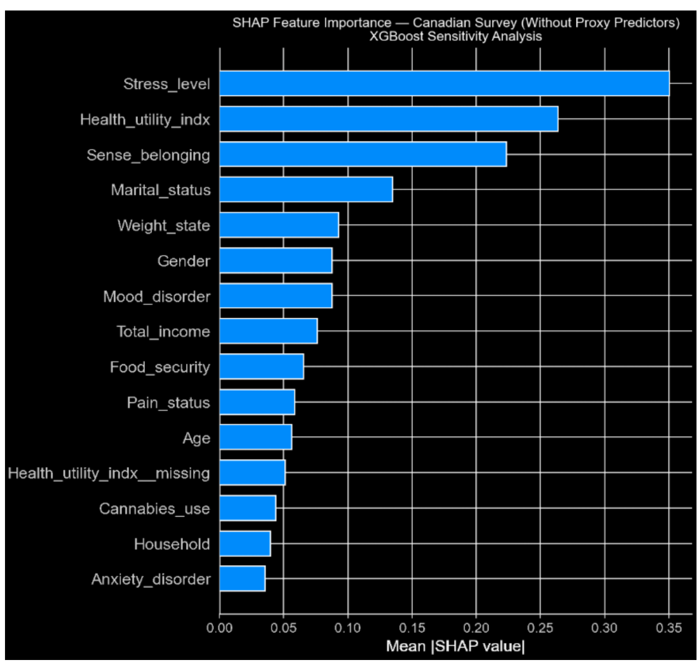

Figure 4.13 - SHAP Bar Plot: Sensitivity Analysis without
Proxy Predictors for Canadian Survey Dataset

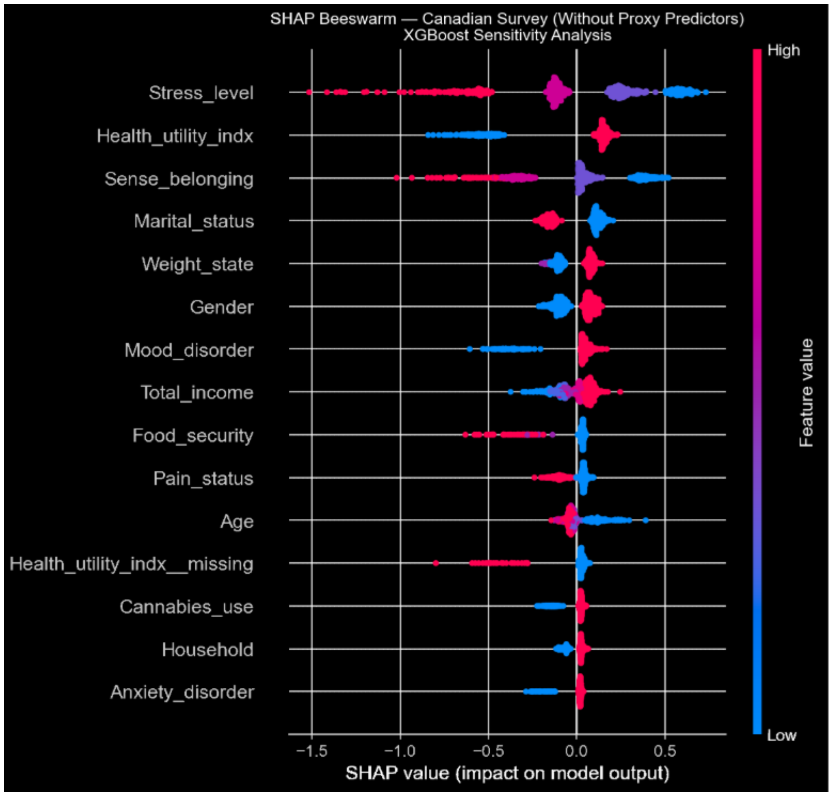

Figure 4.14 - SHAP Beeswarm: Sensitivity Analysis without Proxy Predictors for Canadian Survey Dataset

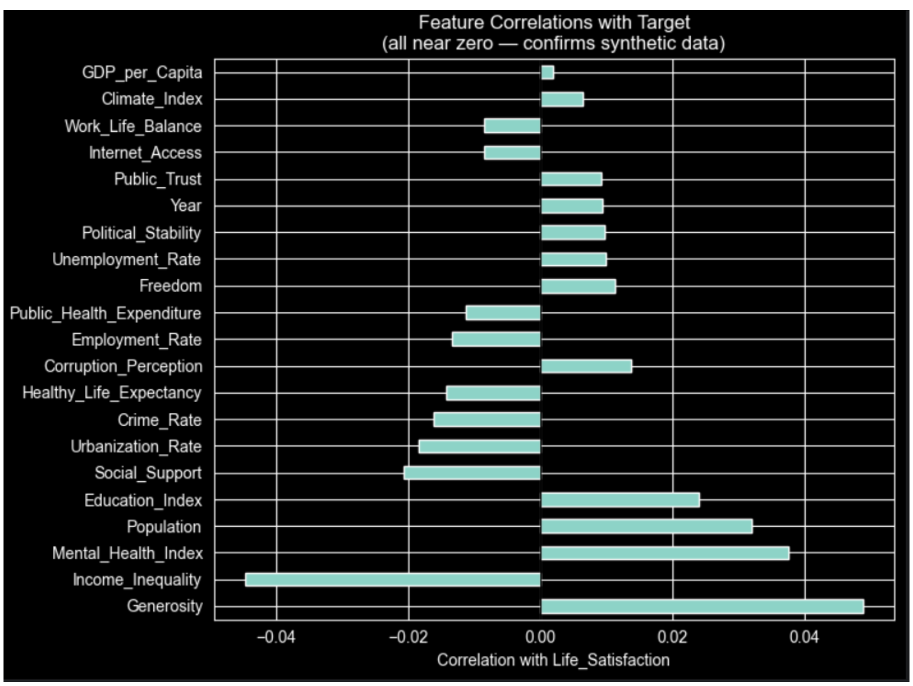

Figure 4.15 - Feature Correlation Chart for Synthetic
World Happiness Report Dataset

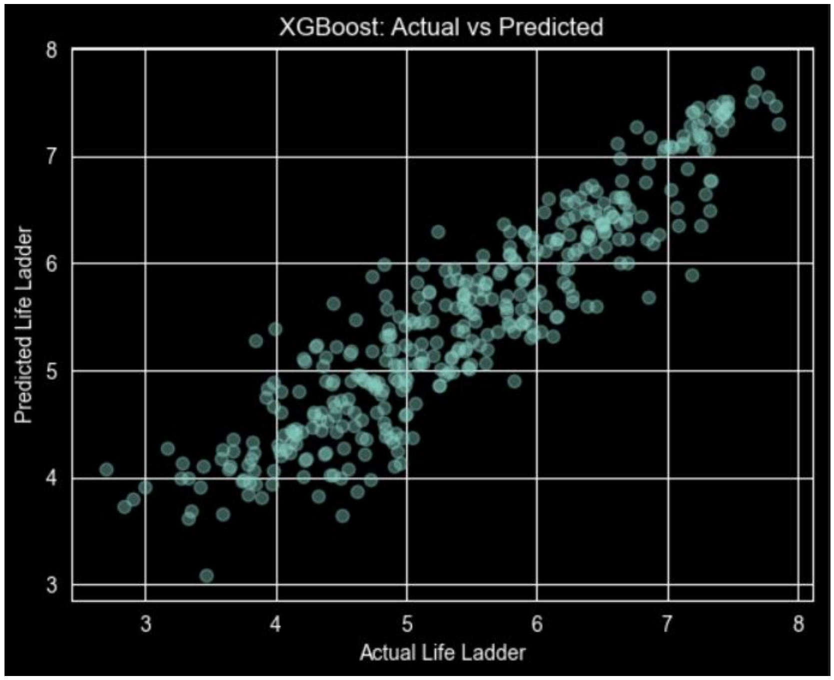

Figure 4.16 - Scatter Plot: XGBoost actual vs predicted
For Gallup World Happiness Report Dataset

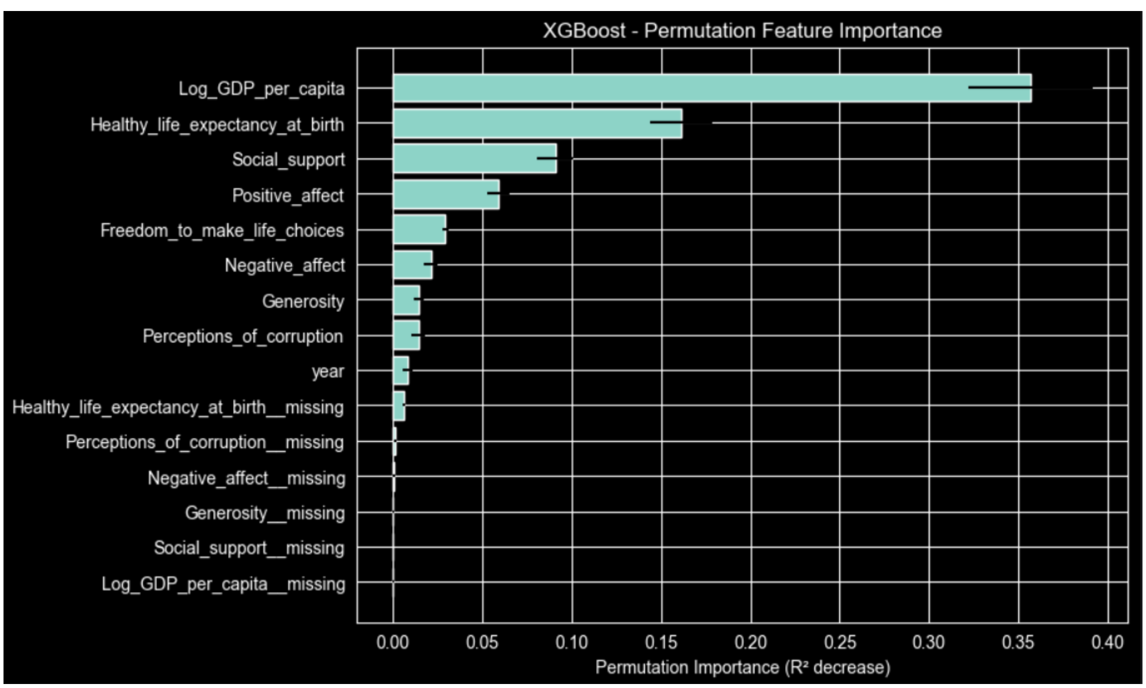

Figure 4.17 - XGBoost Permutation Feature Importance
for Gallup World Happiness Report Dataset

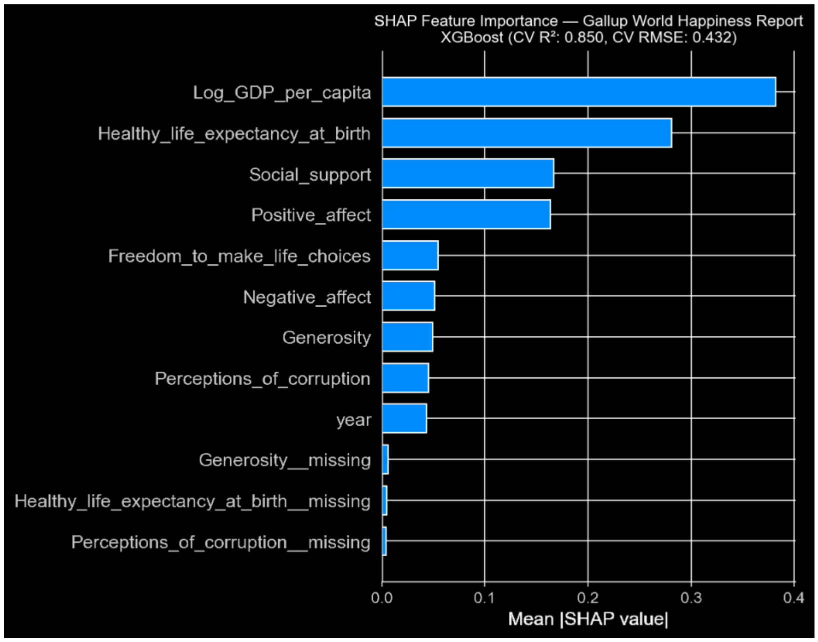

Figure 4.18 - SHAP Bar Plot: Full Model
for Gallup World Happiness Report Dataset

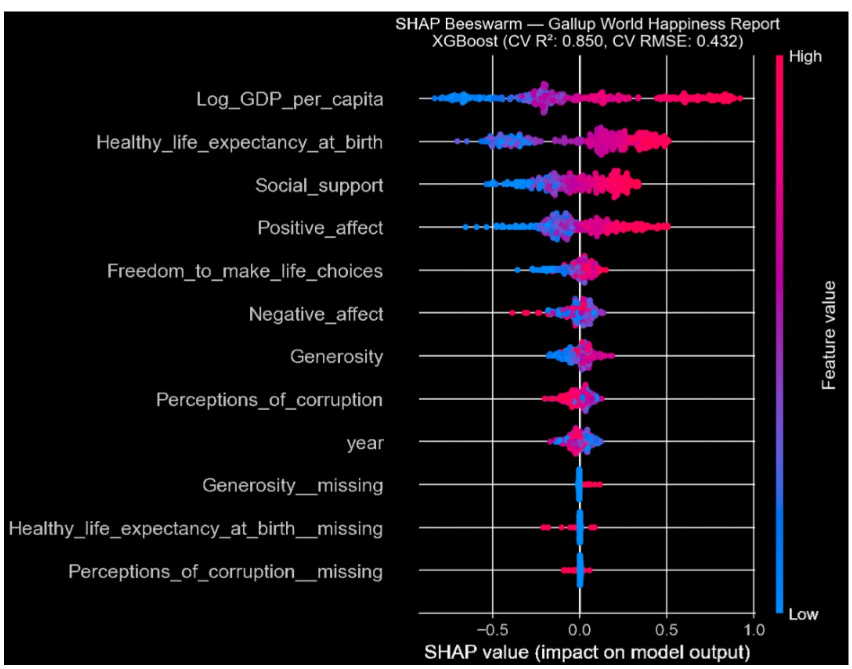

Figure 4.19 - SHAP Beeswarm: Full Model

for Gallup World Happiness Report Dataset

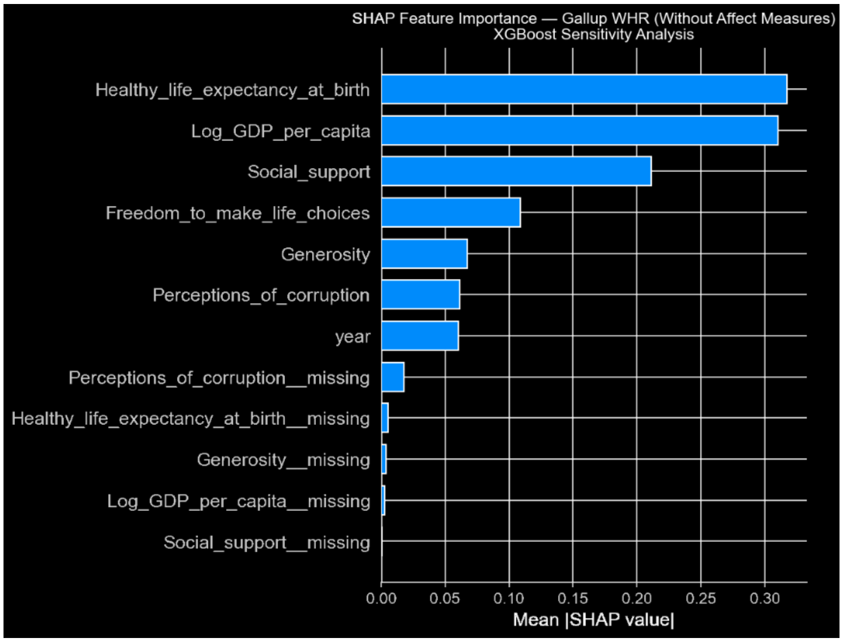

Figure 4.20 - SHAP Bar Plot: Sensitivity Analysis
for Gallup World Happiness Report Dataset

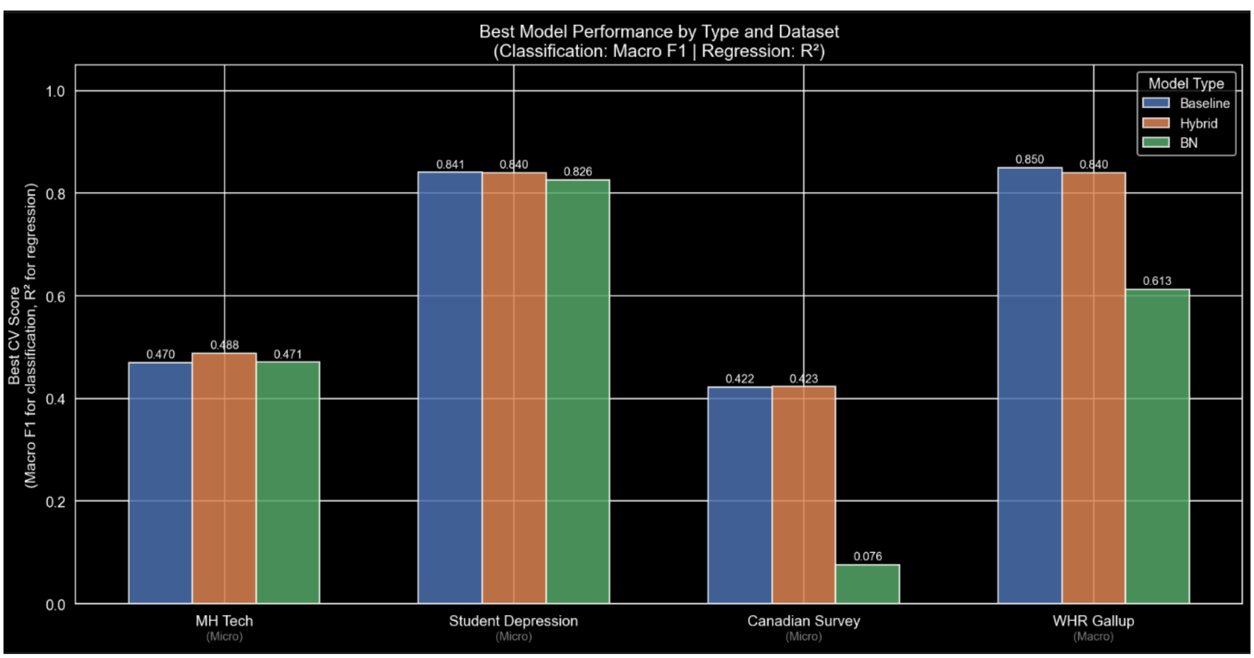

Figure 5.1 - Best Model Performance by Type and Dataset:

Grouped Bar Chart

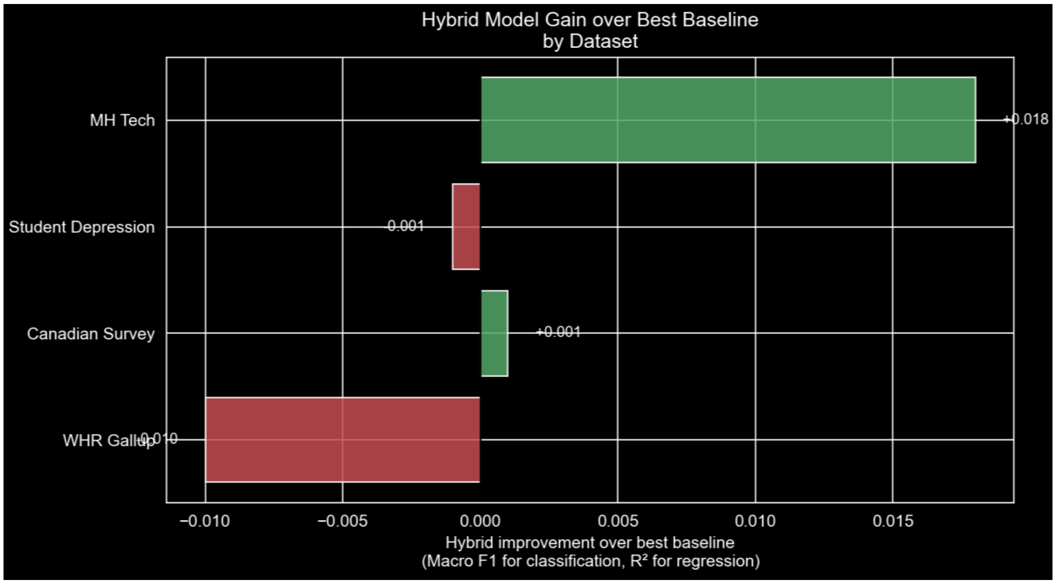

Figure 5.2 - Hybrid Model Gain over Baseline:
Horizontal Bar Chart

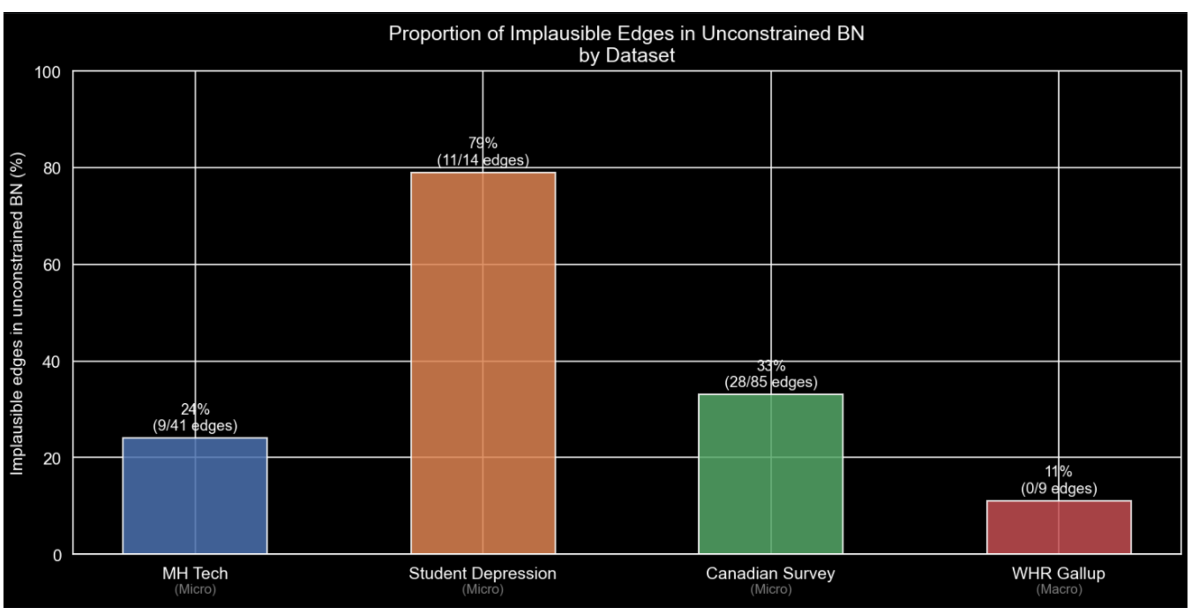

Figure 5.3 - Proportion of implausible edges in unconstrained BN by dataset

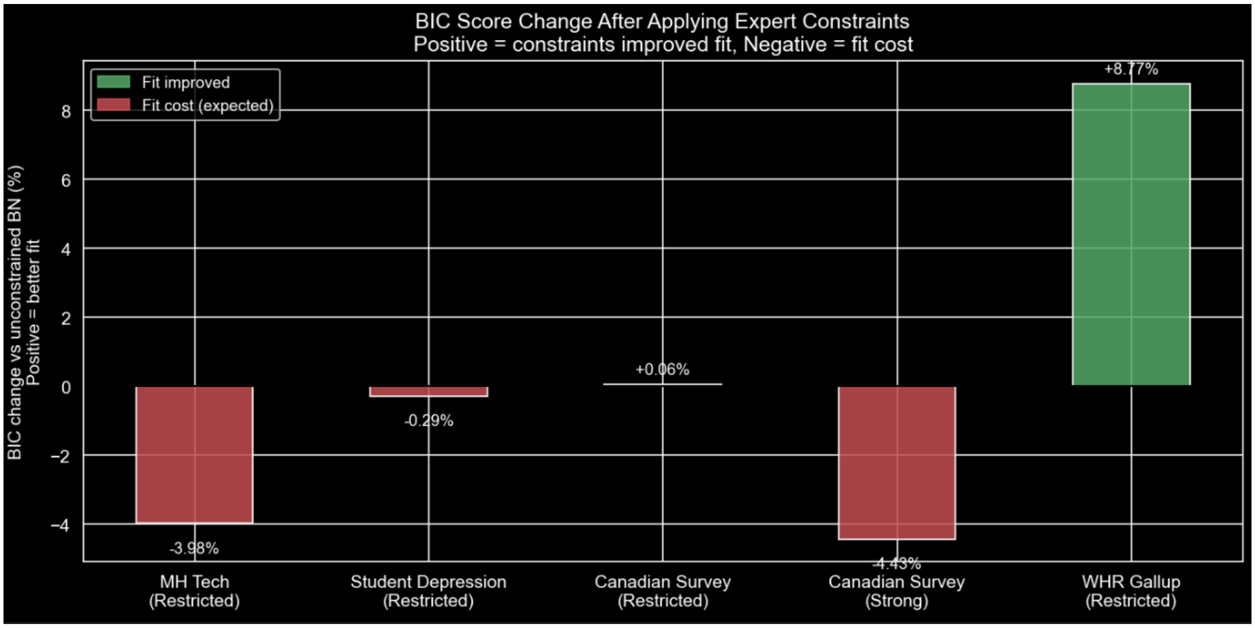

Figure 5.4 - BIC score change after expert constraints by dataset

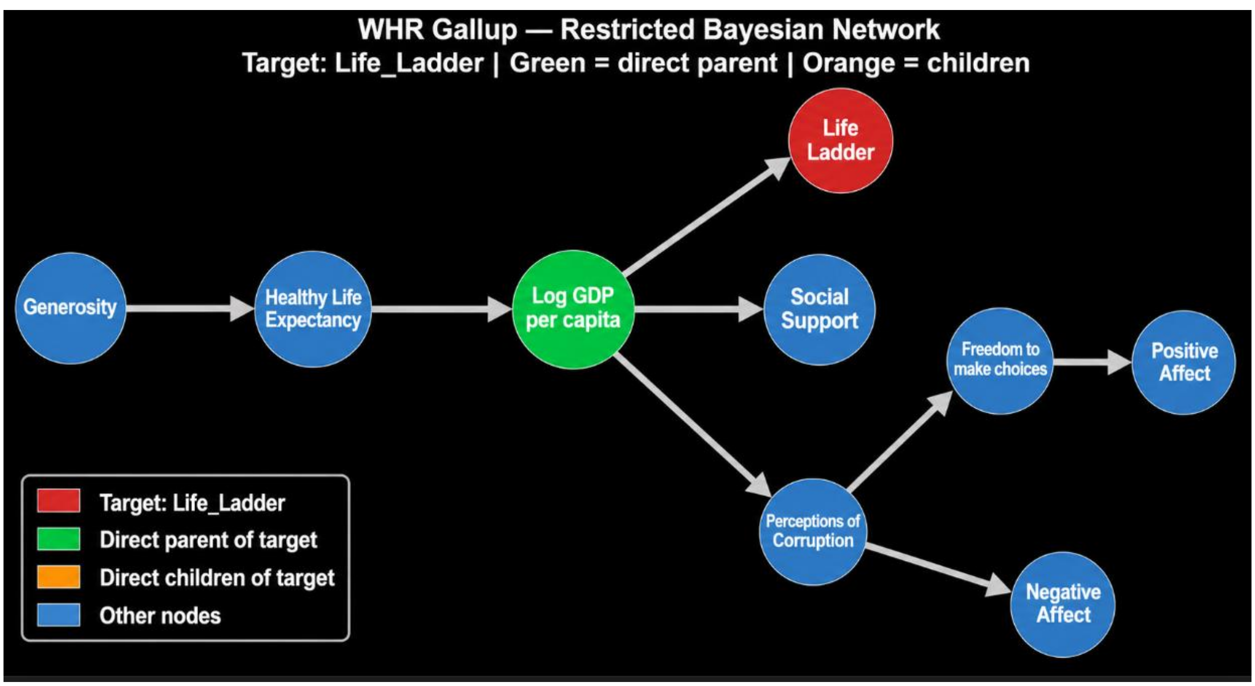

Figure 5.5 - WHR Gallup restricted BN graph (full 9-node structure)

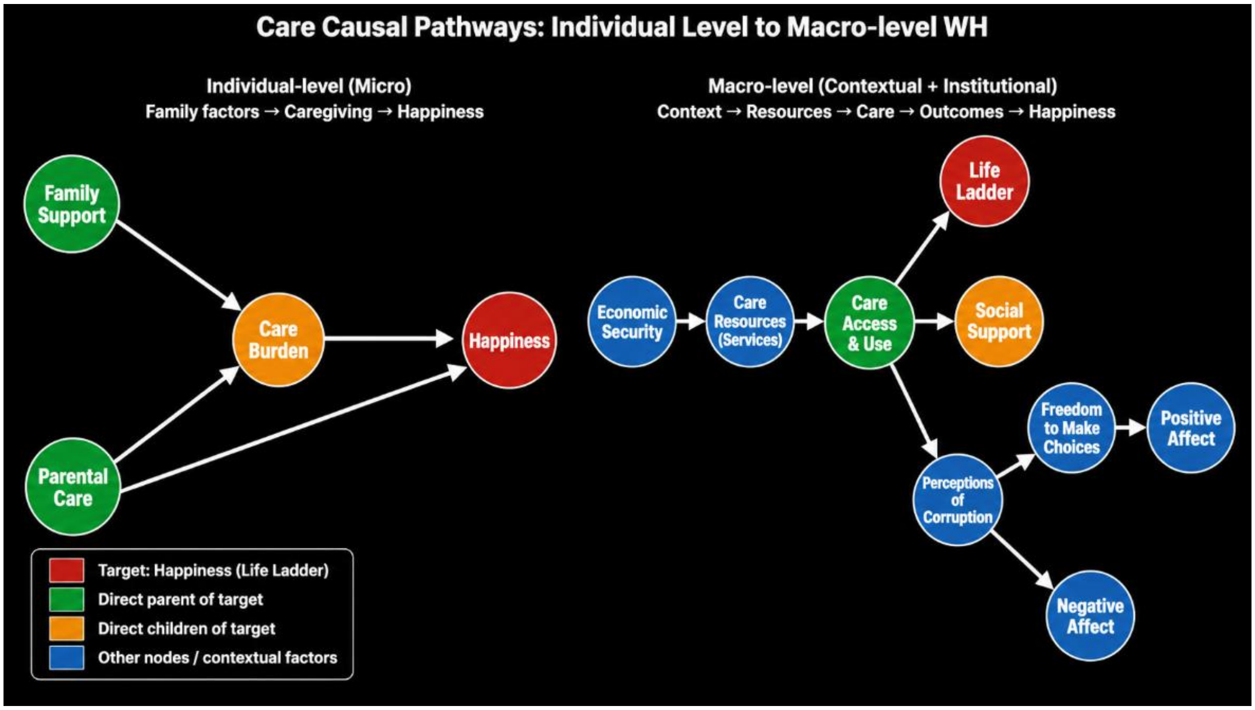

Figure 5.6 - Core causal pathways: individual vs macro level (side-by-side)